MOHAMMED VI
FACULTY OF MEDICINE
UM6SS - CASABLANCA

ANATOMY OF CEREBELLUM

Prof. F. FLIYOU

Module: Anatomy3

Sub-Module: Neuroanatomy – Lecture 6
Semester S3

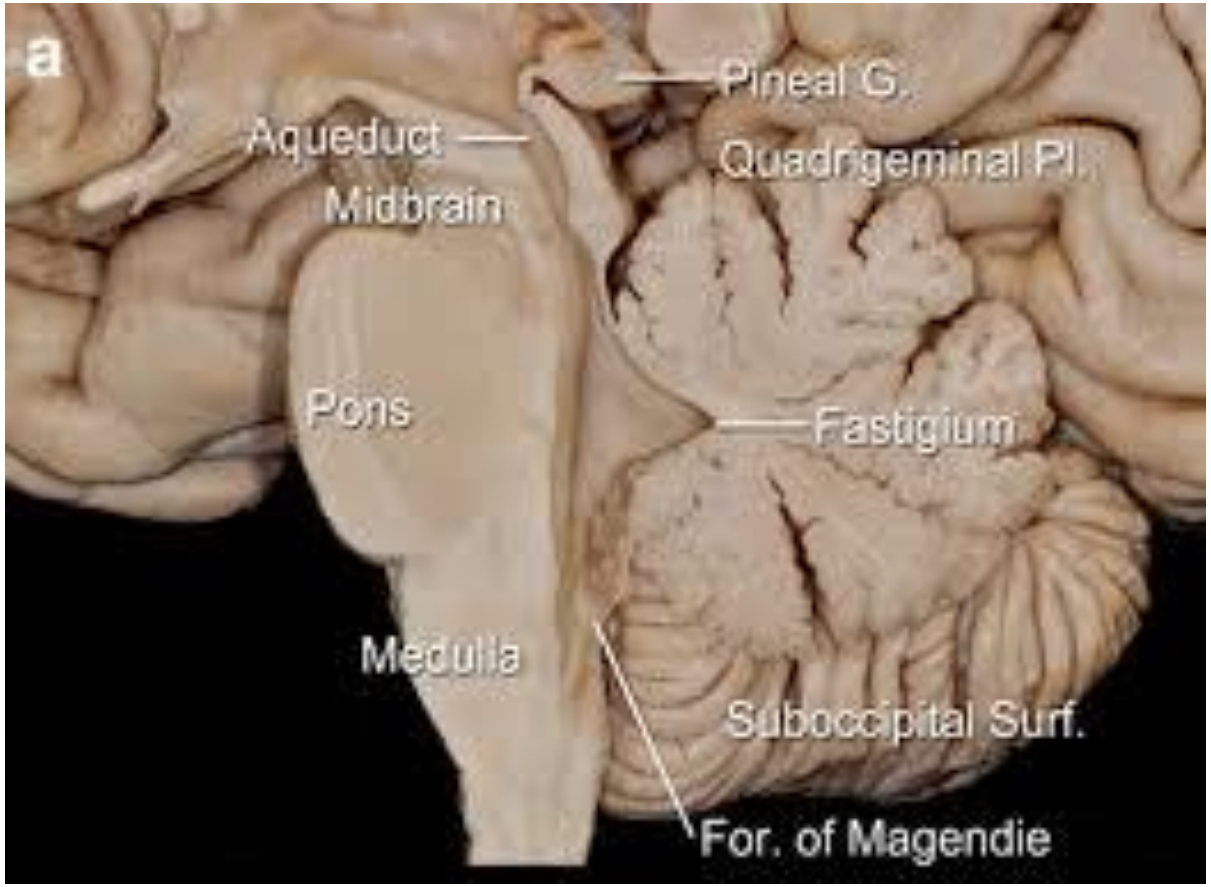
Academic year: 2026-2027

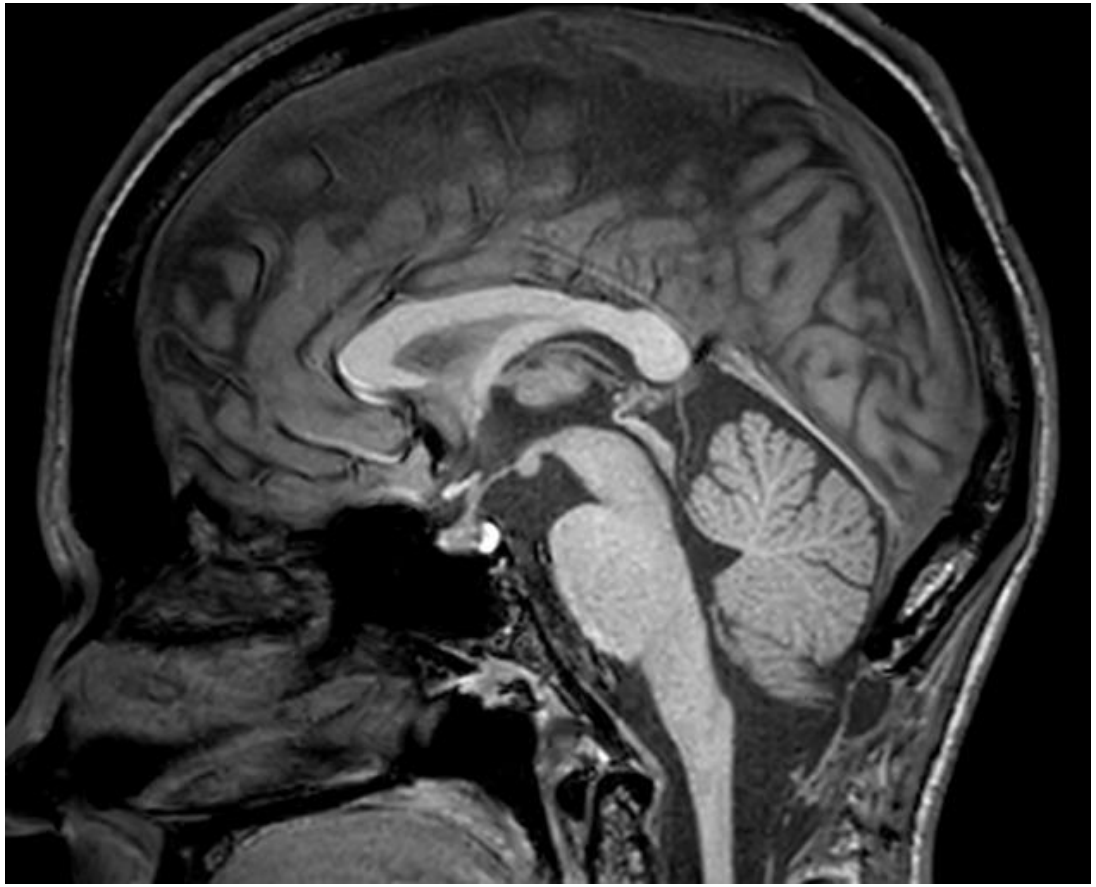
MOHAMMED VI UNIVERSITY
OF SCIENCES AND HEALTH
UM6SS

www.um6ss.ma

Plan:

I. Introduction
II. Embryological review
III. General morphology
IV. External configuration
V. Internal configuration
VI. Functional anatomy
VII. Clinical aspects

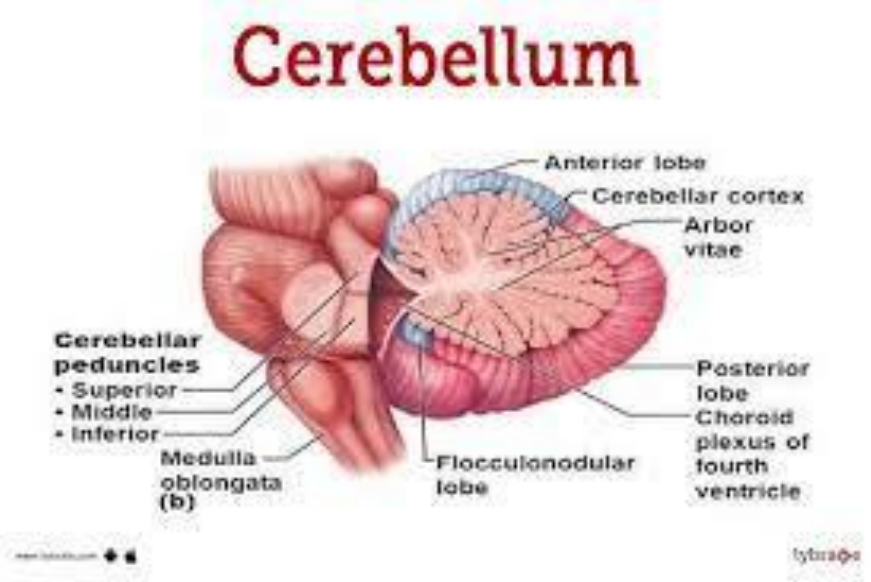

a
Pineal G.
Aqueduct
Quadrigeminal Pl.
Midbrain
Pons
Fastigium
Medulla
Suboccipital Surf.
For. of Magendie

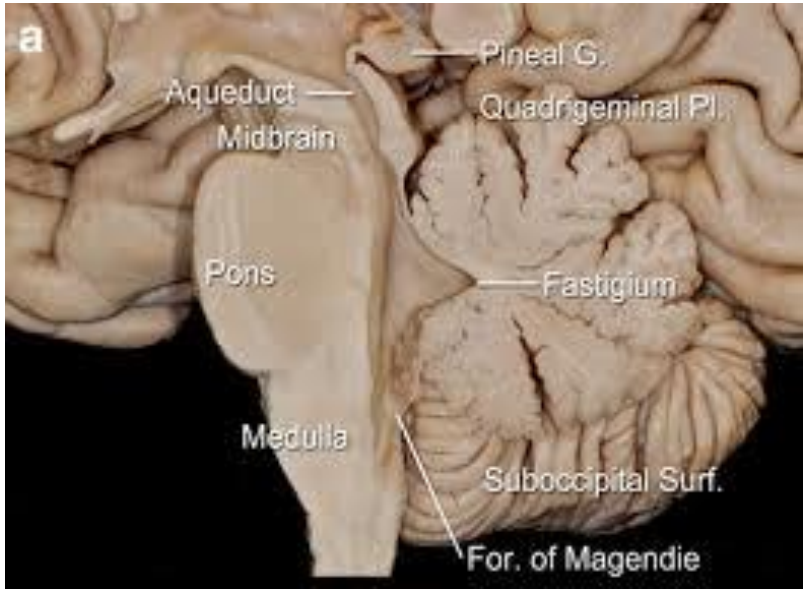




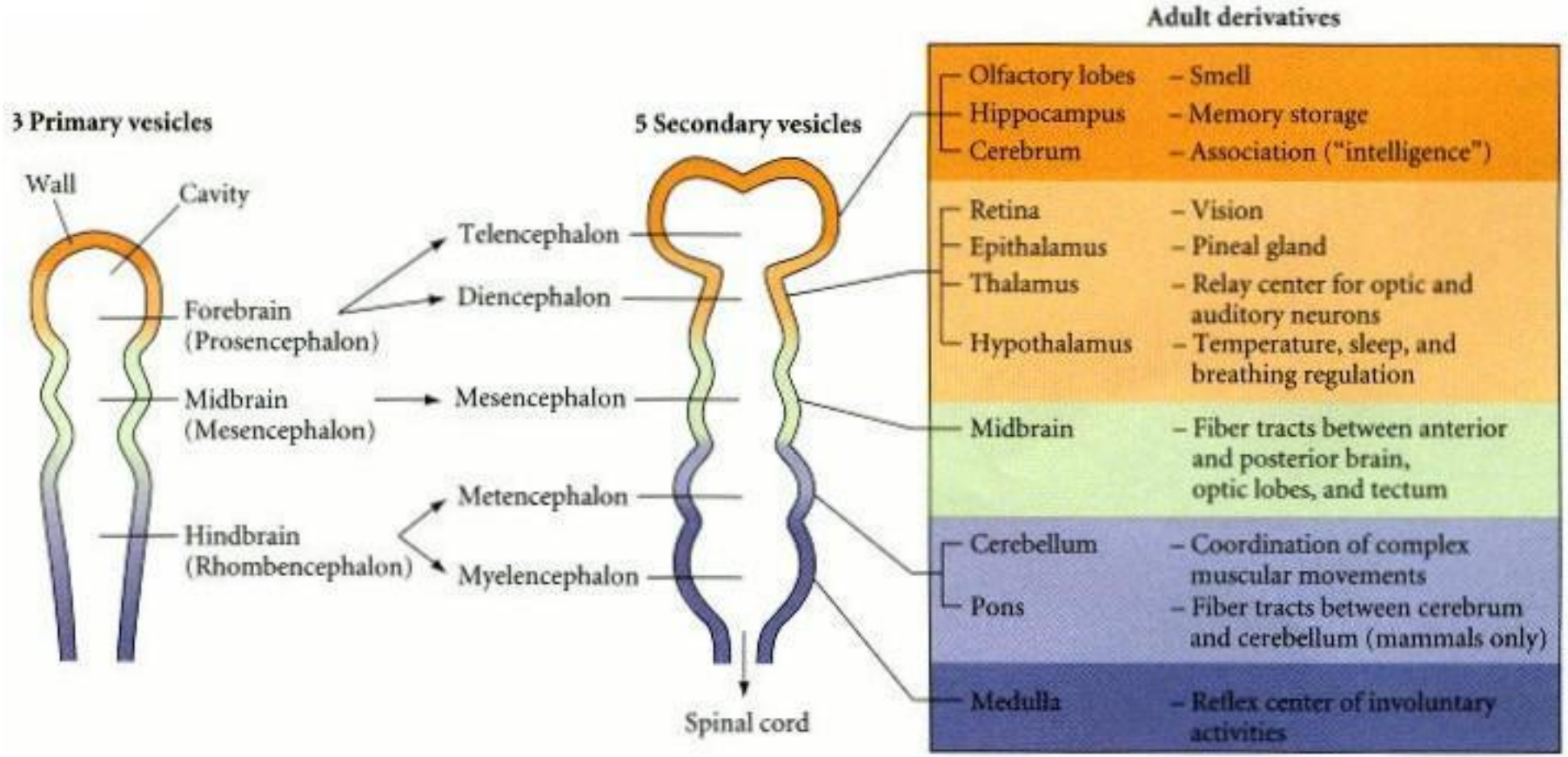
INTRODUCTION

- The cerebellum is a part of CNS that located in the posterior cerebral fossa:
  - ➢inferior to the tentorium cerebelli
  - ➢behind the pons and medulla oblongata
  - ➢separated from these structures by cavity of fourth ventricle.
- It is connected to brainstem by three fibre tracts known as cerebellar peduncles
- It plays a crucial role in the automatic regulation of motor activity, including the control of muscle tone, the maintenance of balance, and the coordination of voluntary movements.





EMBRYOLOGICAL RECALL

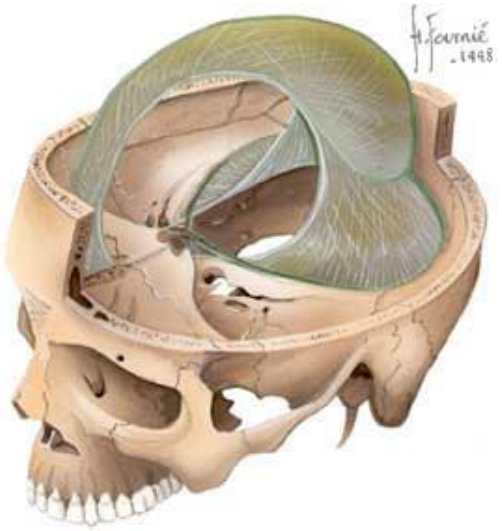


External configuration

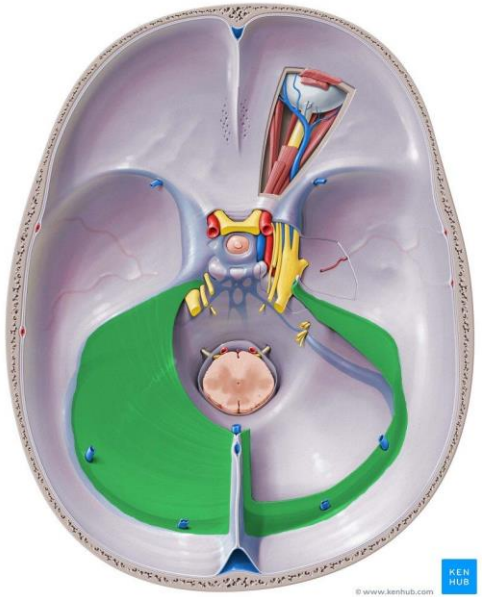
- **Location**: Posterior cranial fossa, behind brainstem
- **Divisions**:

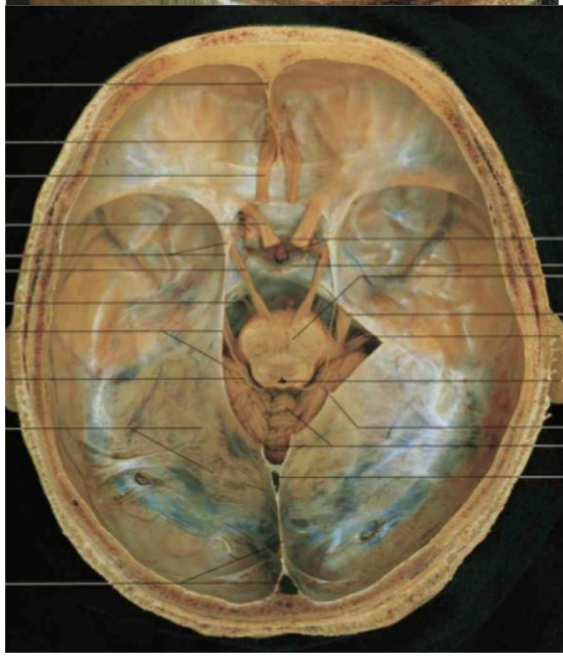
Cerebellum consists of two large cerebellar hemispheres united to each other by a median portion, vermis.

- **Color**: Grayish-pink



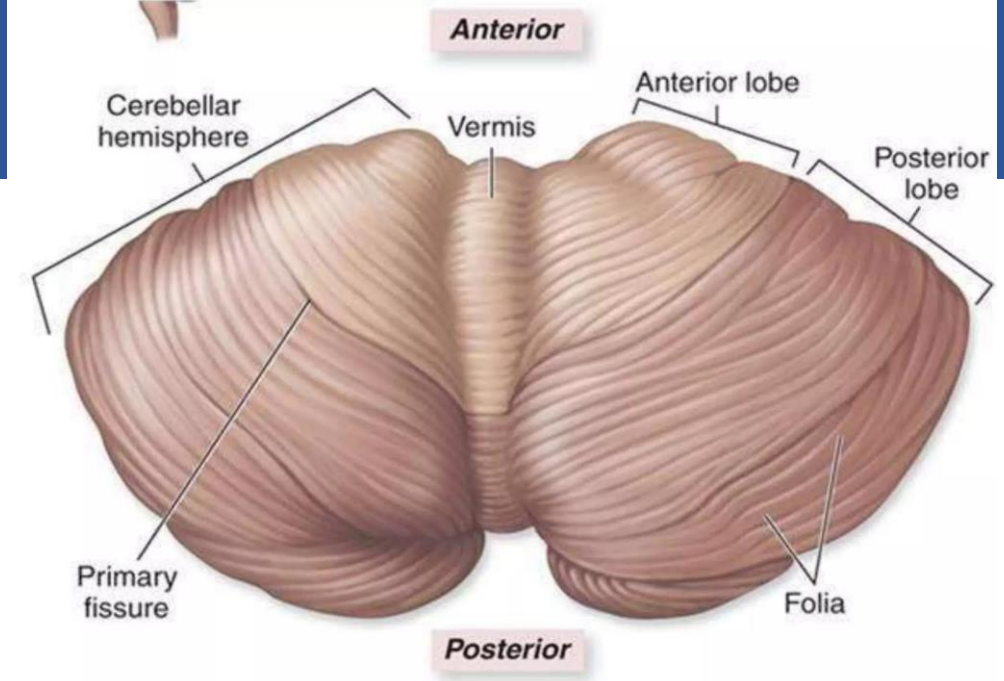




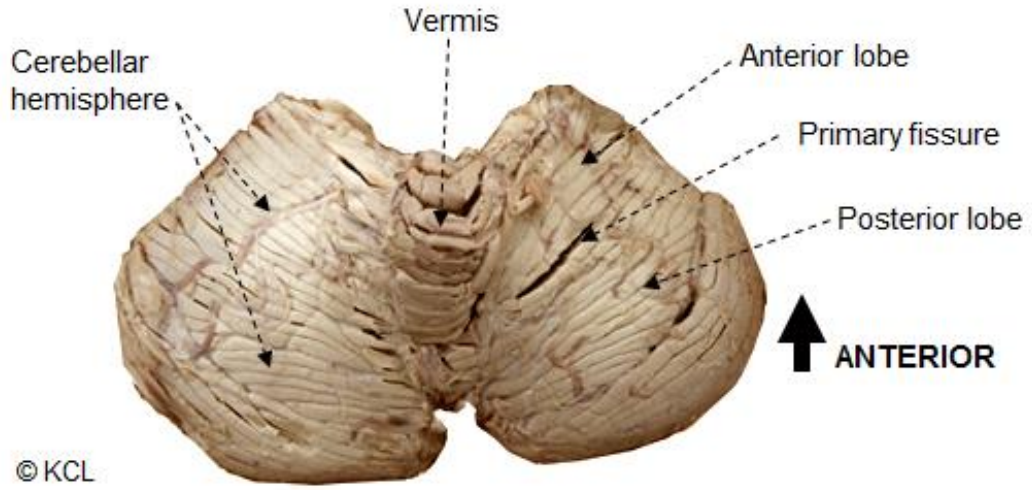


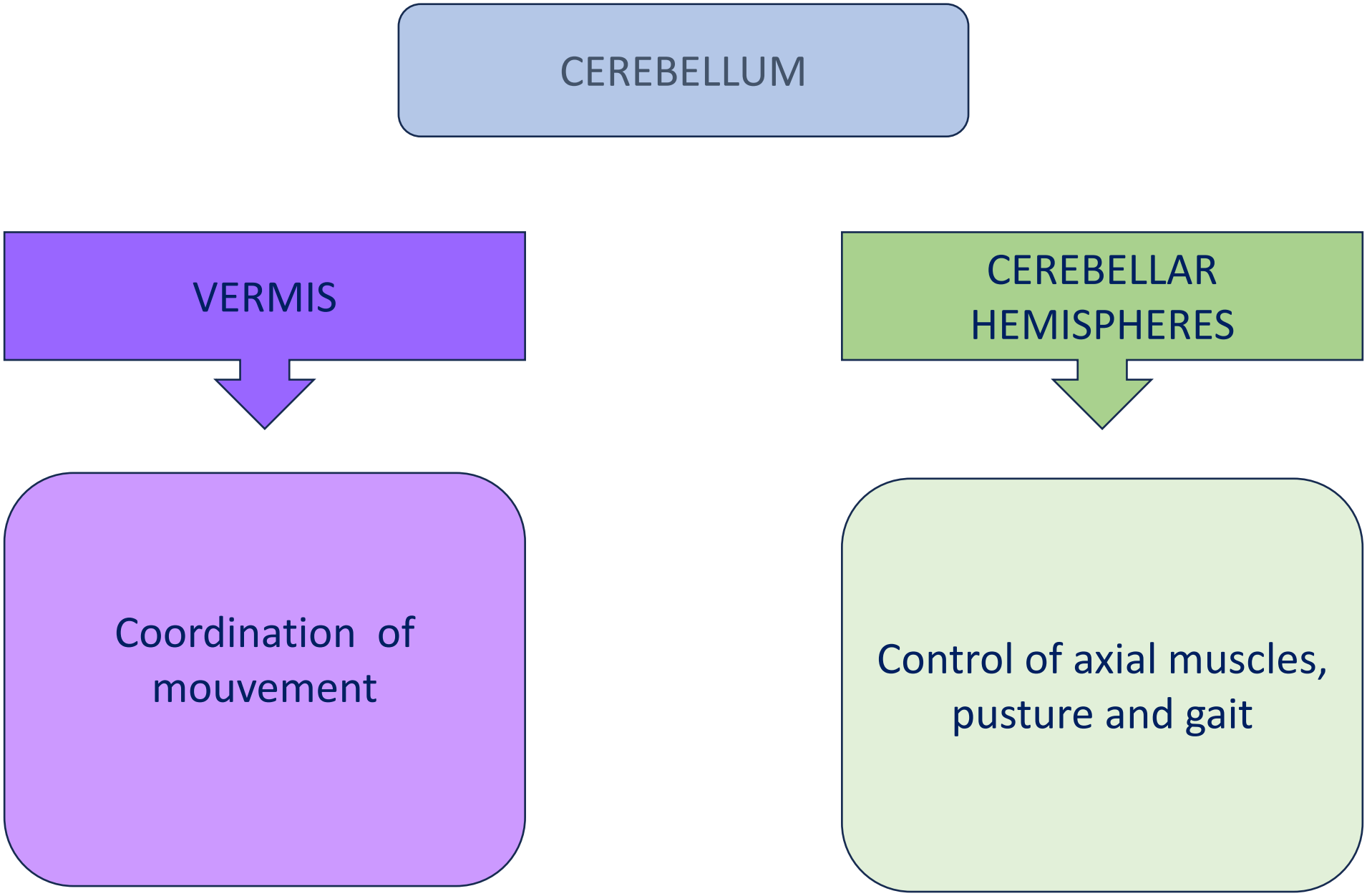
General morphology

- Cerebellum consists of two large bilateral lobes called cerebellar hemispheres. These two lobes are united to each other by a median portion, vermis.



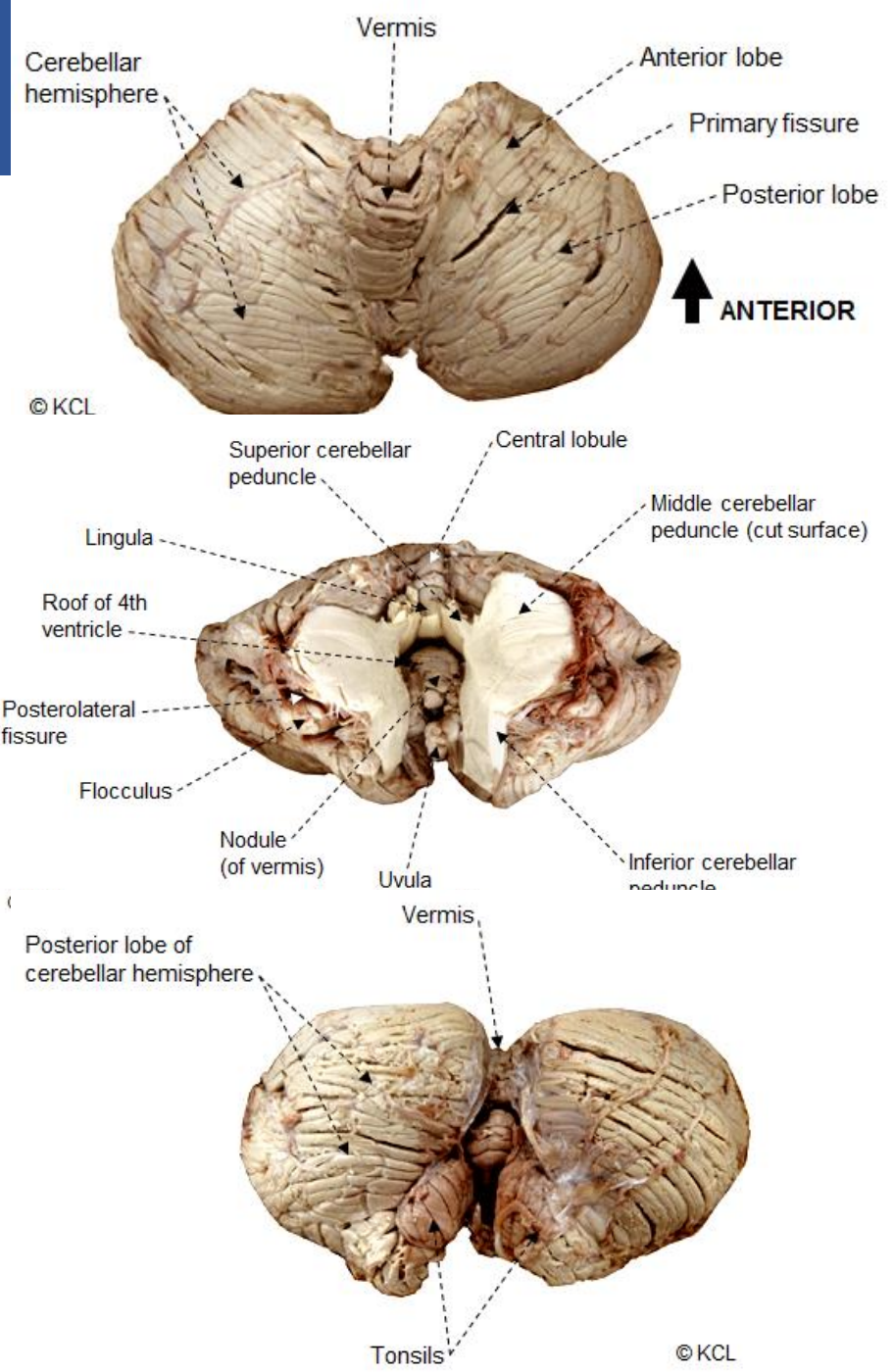
(b) Cerebellum, superior view




CEREBELLUM
VERMIS
CEREBELLAR HEMISPHERES
Coordination of mouvement
Control of axial muscles, pusture and gait

General morphology

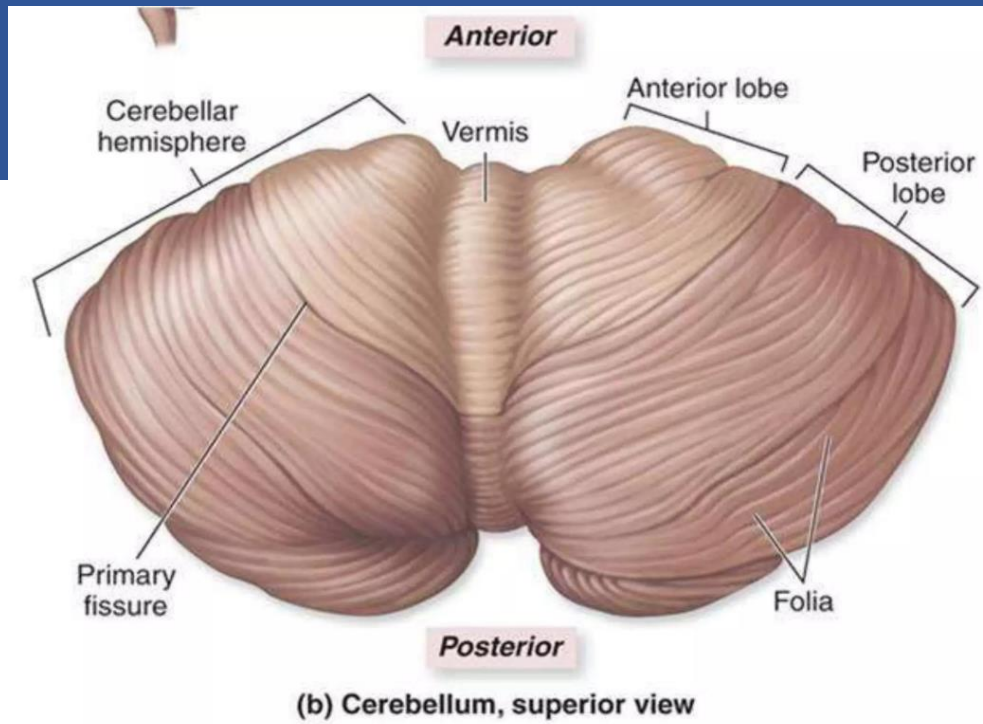
- Cerebellar surface is divided by numerous curve transverse giving it a laminated appearance
- Grossly cerebellum comprises of three parts: two surfaces, two notches and three well demarcated fissures



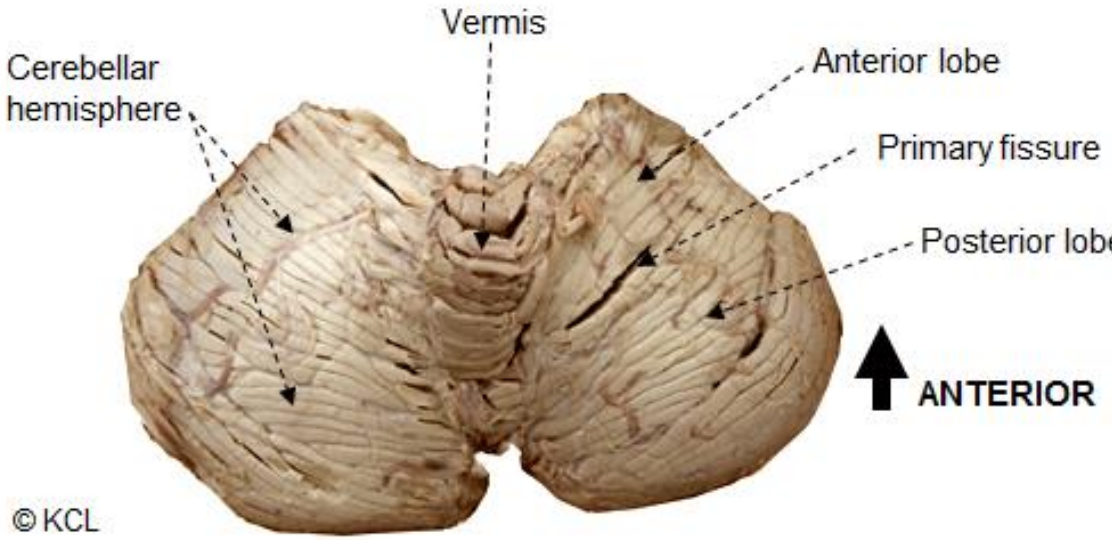
External configuration

❖**Surfaces**:It presents two main **surfaces**:

- A **superior surface**, convex and related to the tentorium cerebelli.



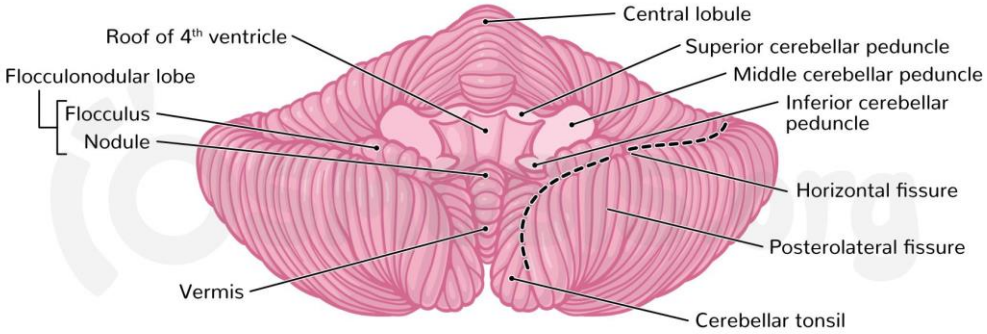
(b) Cerebellum, superior view

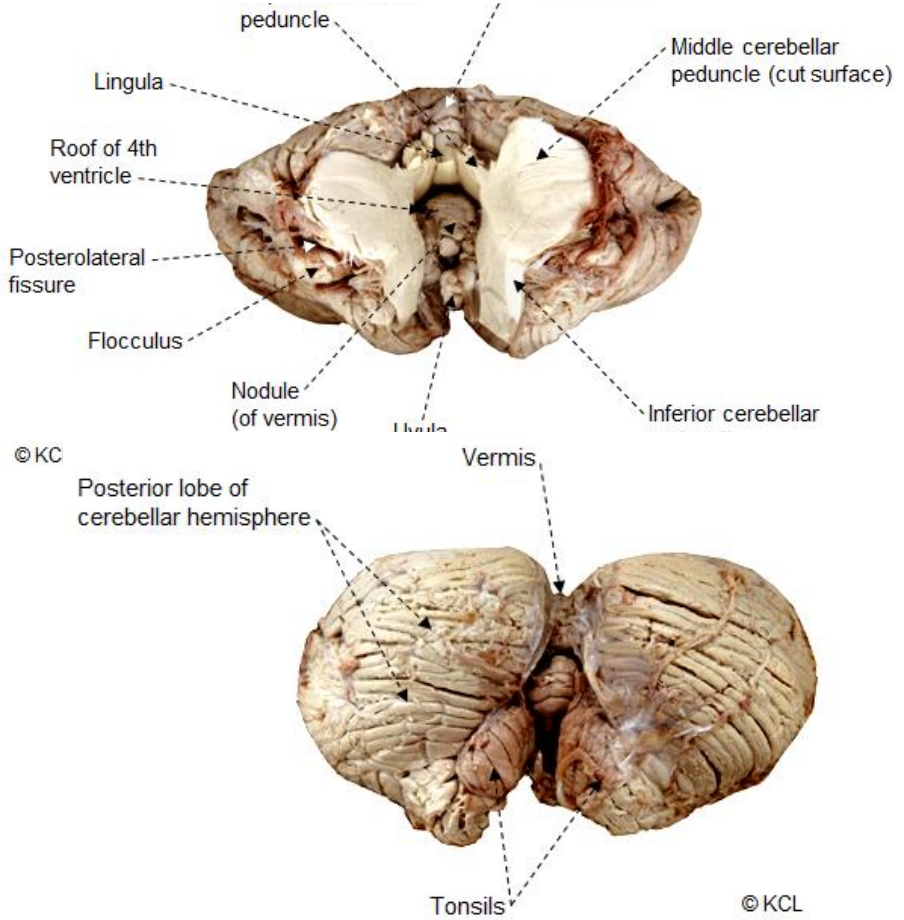


External configuration

❖Surfaces:

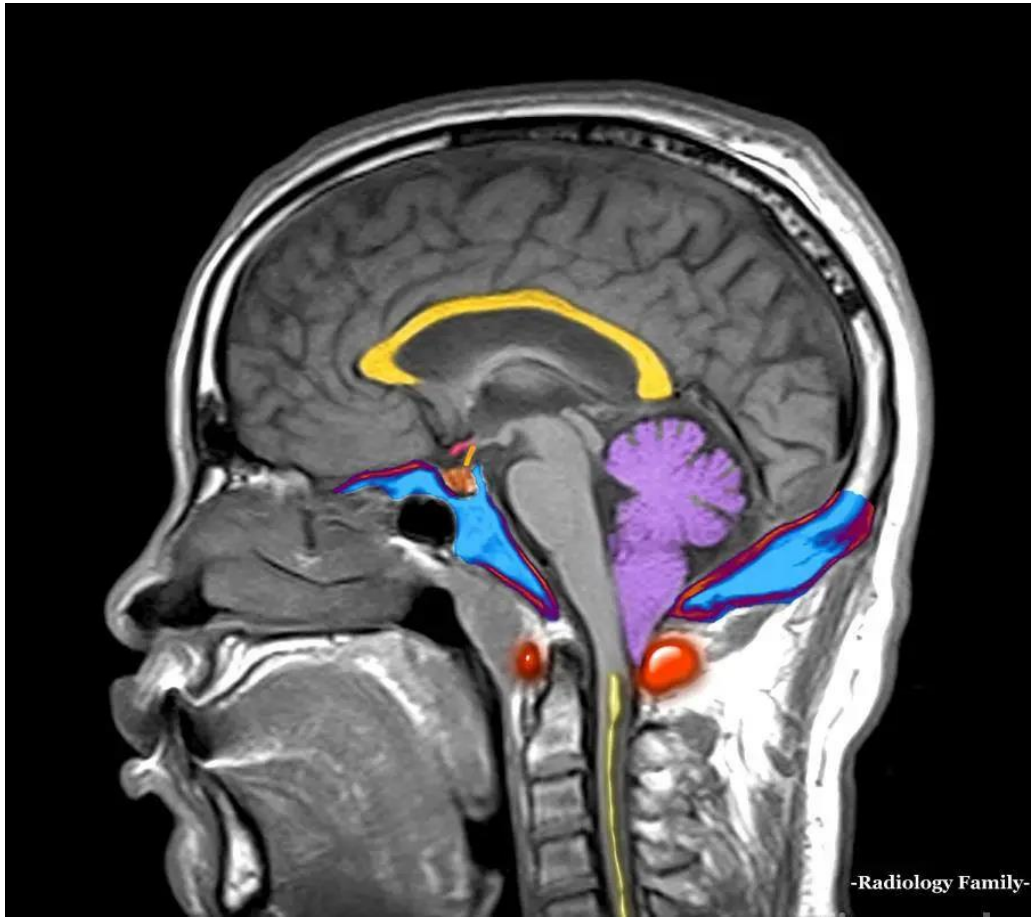
- An **inferior surface**, adapted to the posterior fossa and separated by the **vallecula**, a deep median groove that houses the brainstem.
- The cerebellar tonsils: rounded portions of medial inferior cerebellar hemispheres, on either side of inferior vermis





Cerebellar tonsillar herniation

Compress the medulle oblongata
Threatening vital respiratory and cardiovascular functions

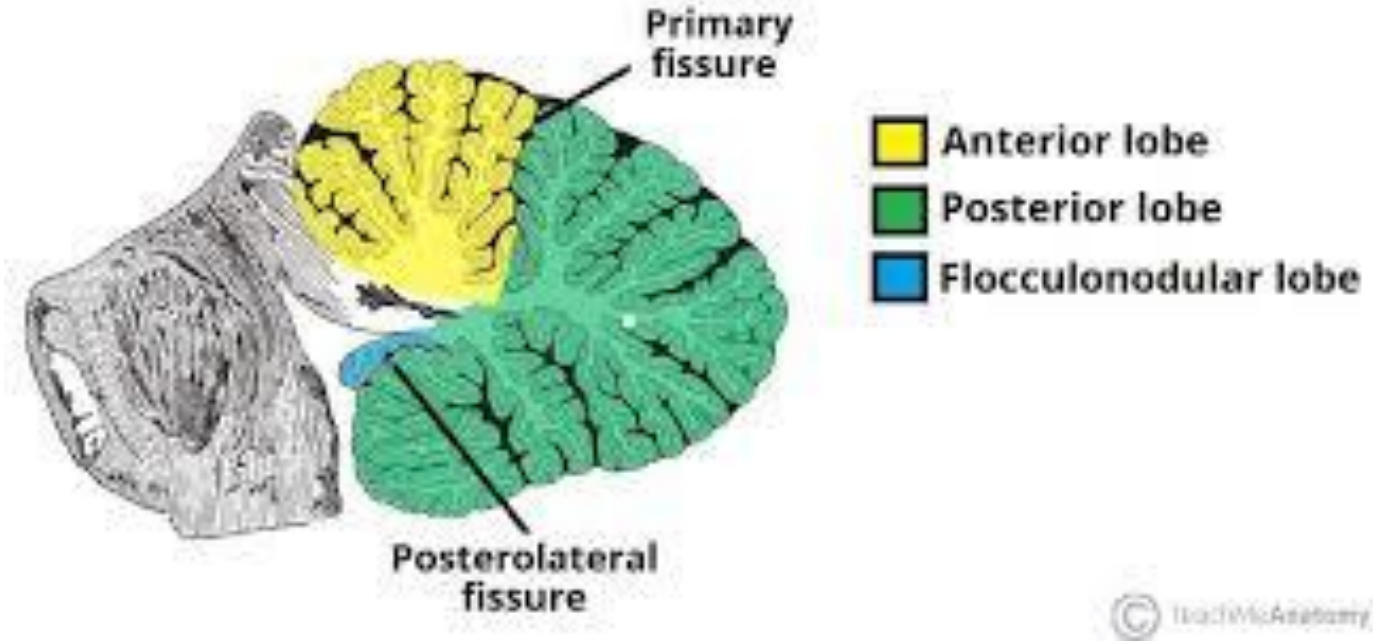


Descriptive anatomy: External configuration

❖**Furrows**: First, second- and third-order sulci

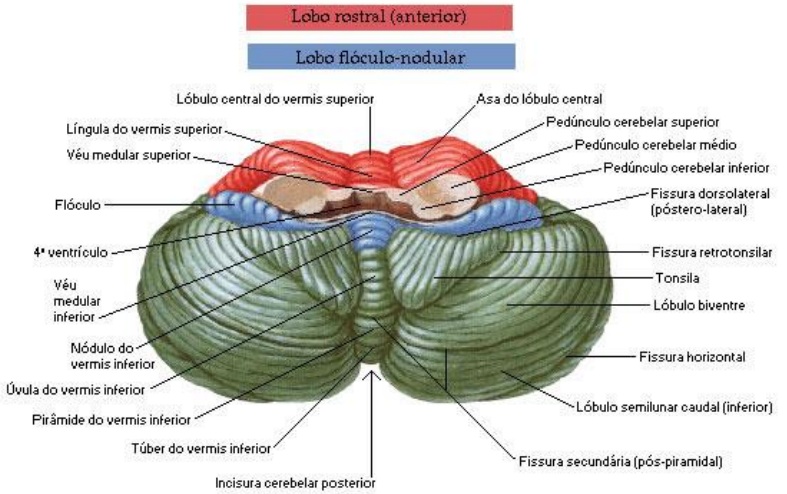
Some of the fissures are used as landmarks to divide the cerebellum into **lobes**:

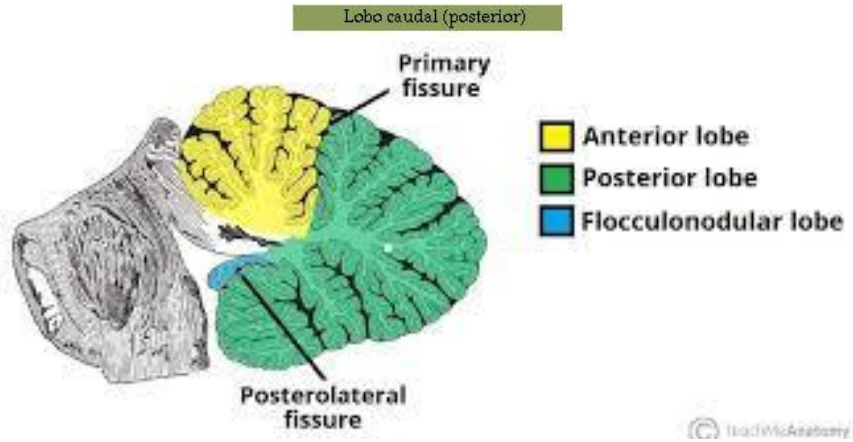
- The **primary fissure** divides the smaller anterior and larger posterior lobes
- The **posterolateral fissure** demarcates the flocculonodular and posterior lobes.
- The most important of these is the **great circumferential sulcus of Vicq d'Azyr (horizontal fissure)**, which separates the **superior surface** from the **inferior surface**.

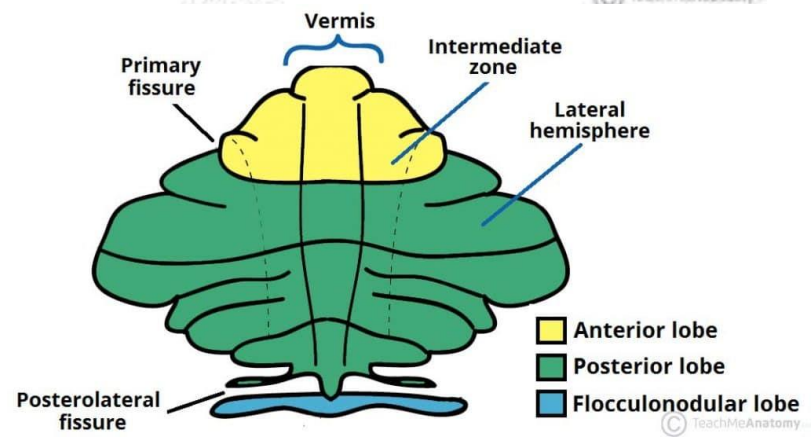


Descriptive anatomy: External configuration

- There are three anatomical lobes that can be distinguished in the cerebellum. These lobes are divided by two fissures – the primary fissure and posterolateral fissure
  - ➢The anterior lobe
  - ➢The posterior lobe
  - ➢The flocculonodular lobe. It is the oldest part of the brain in evolutionary terms (archicerebellum) and participates mainly in balance and spatial orientation. Its primary connections are with the vestibular nuclei, although it also receives visual and other sensory input.





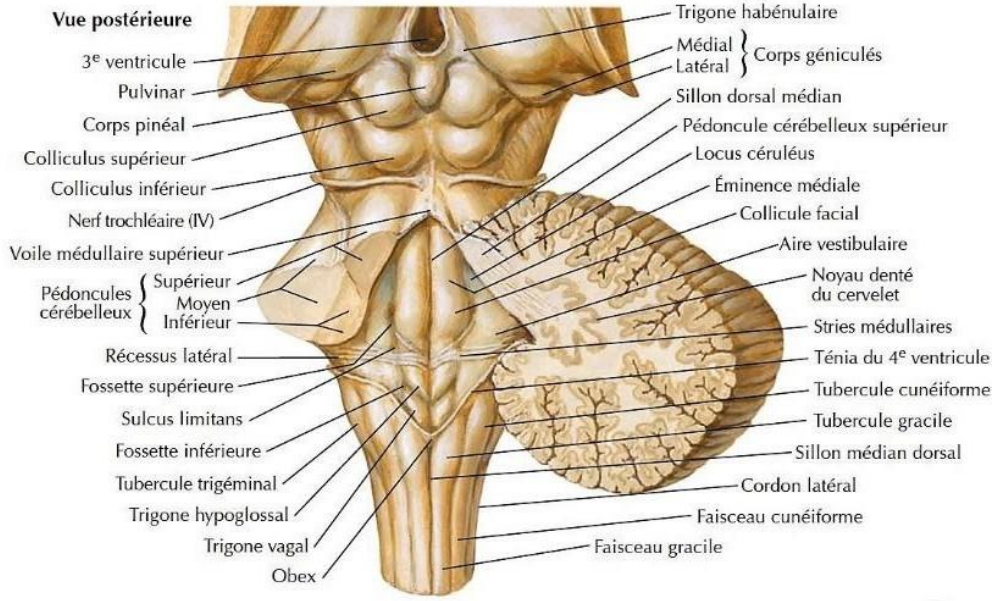


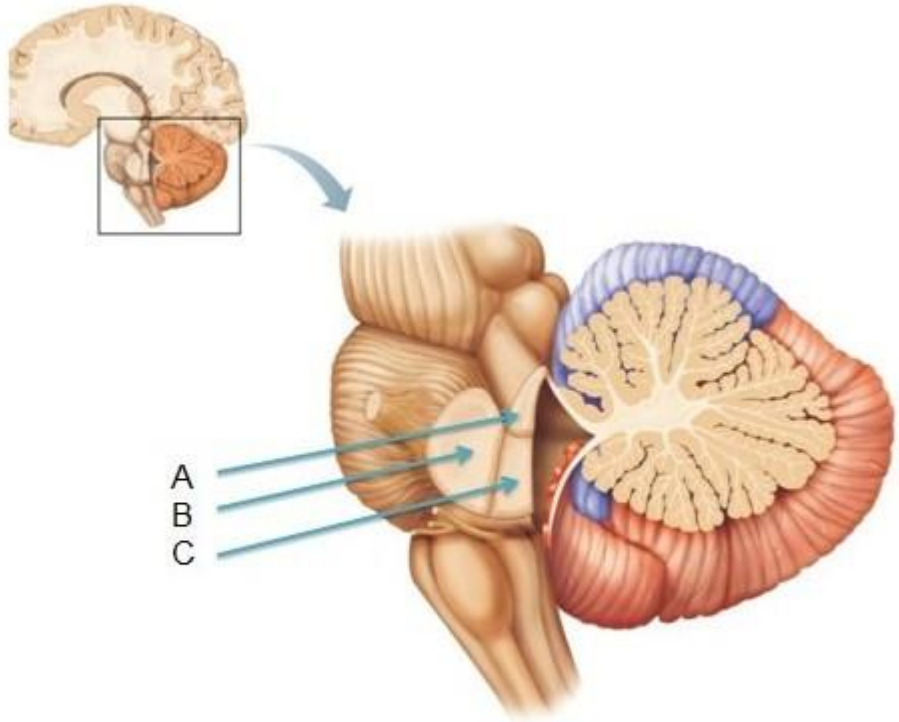
Descriptive anatomy: External configuration

Cerebellum is connected to brainstem by three large bundles of fibres called cerebellar peduncles.

- Superior peduncle connects cerebellum with midbrain
- middle with pons
- inferior with medulla oblongata

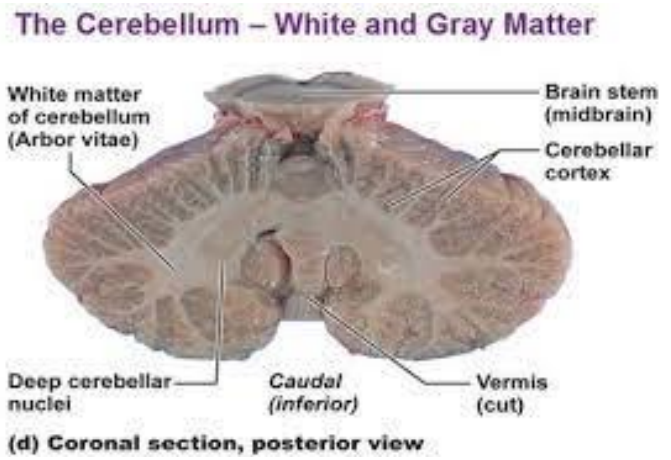
The afferent and efferent fibers pass through these peduncles.

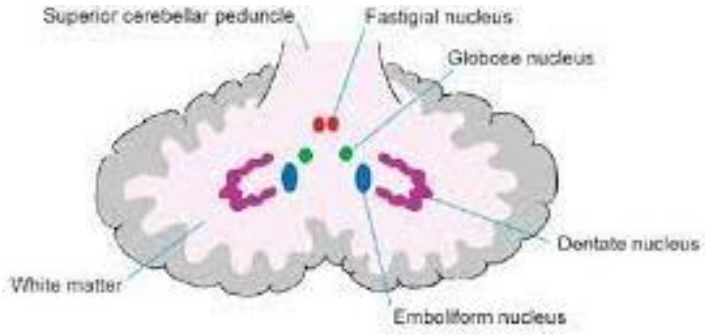




Descriptive anatomy: Internal configuration

On sectioning, the cerebellum shows a characteristic **tree-like pattern** called the *arbor vitae*, formed by branching **white matter** surrounded by **gray matter** of the cortex.

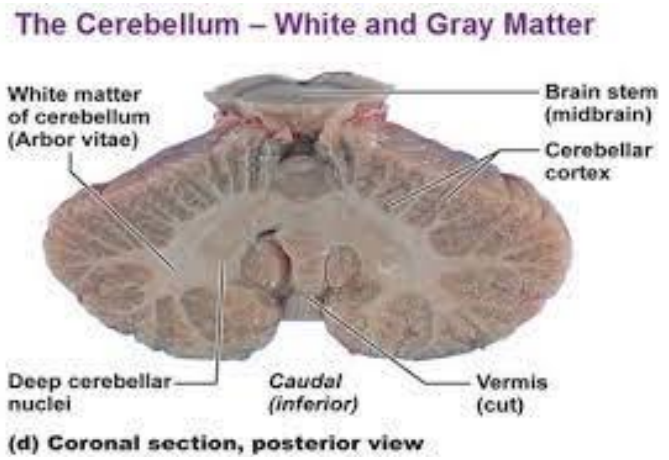


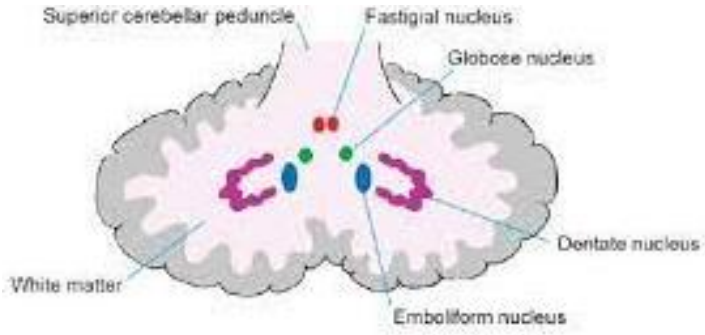


Descriptive anatomy: Internal configuration

The cerebellum consists of grey matter and white matter:

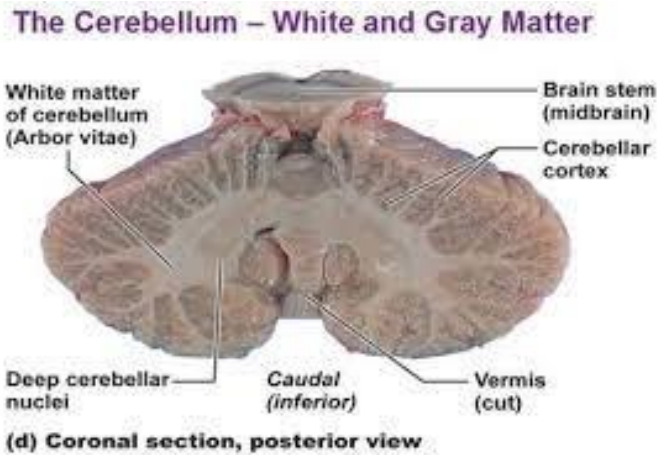
- Grey matter
- White matter

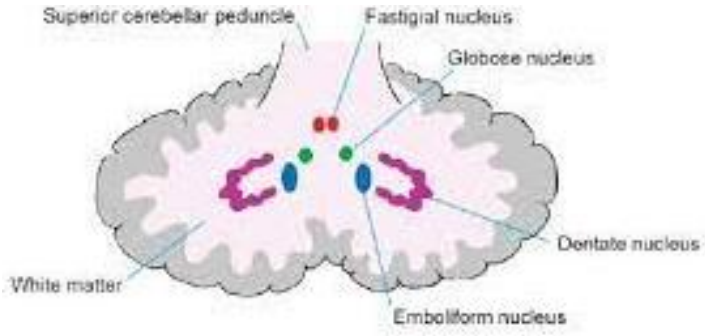




Descriptive anatomy: Internal configuration

- Grey matter located:
  - ➢on the surface of the cerebellum. It is tightly folded, forming the cerebellar cortex.
  - ➢the four cerebellar nuclei:
    - ✓the dentate
    - ✓Emboliform
    - ✓Globose
    - ✓fastigial nuclei
  - ➢The interposed nuclei consist of two subnuclei:Globose nucleus + Emboliform nucleus



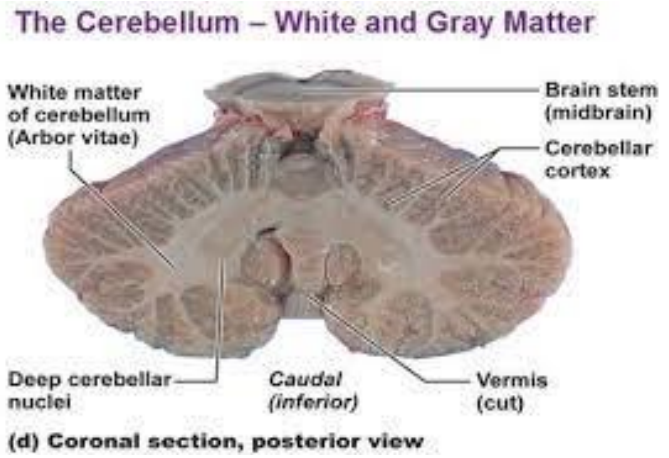


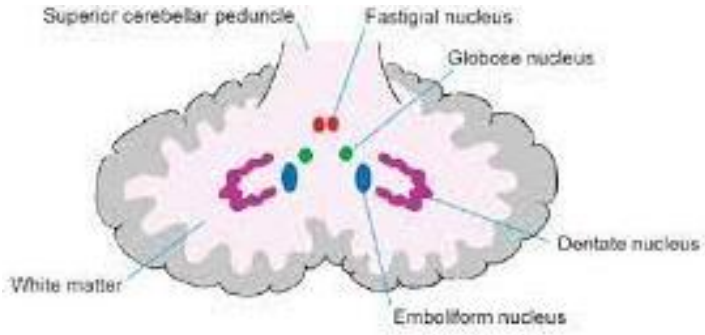
Descriptive anatomy: Internal configuration

- White matter:

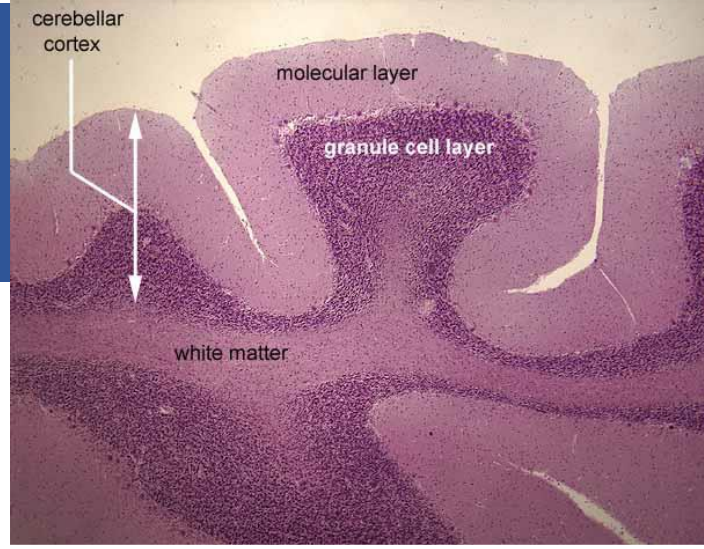
➢ located underneath the cerebellar cortex and between the cerebellar nuclei

➢ forming a characteristic branching pattern known as the **“arbor vitae”** (*tree of life*)



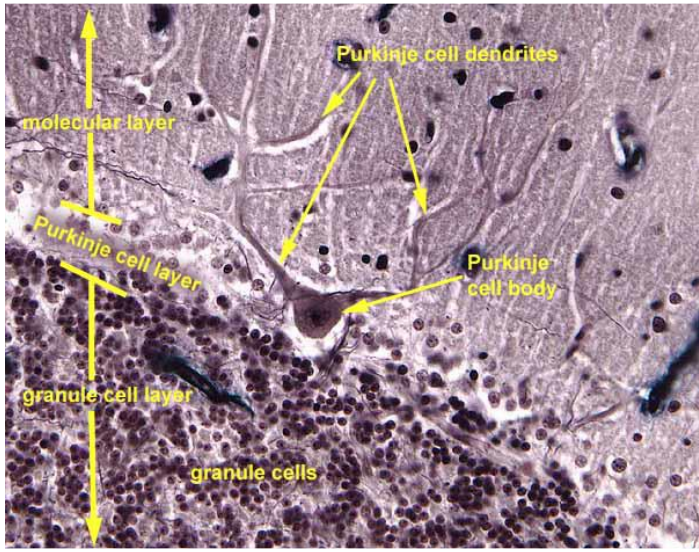


Descriptive anatomy: Internal configuration



The **cerebellar cortex** consists of **three layers**:

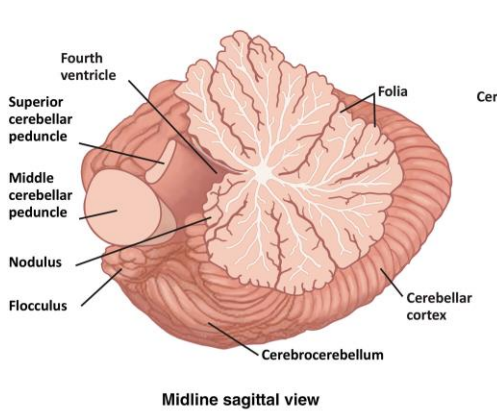
1. **Molecular layer:** contains inhibitory interneurons (stellate and basket cells) and abundant dendritic arborizations.
2. **Purkinje cell layer:** a single row of large Purkinje cells, the only output neurons of the cerebellar cortex, projecting inhibitory signals to the deep nuclei.
3. **Granular layer:** densely packed with small granule cells and inhibitory Golgi cells.



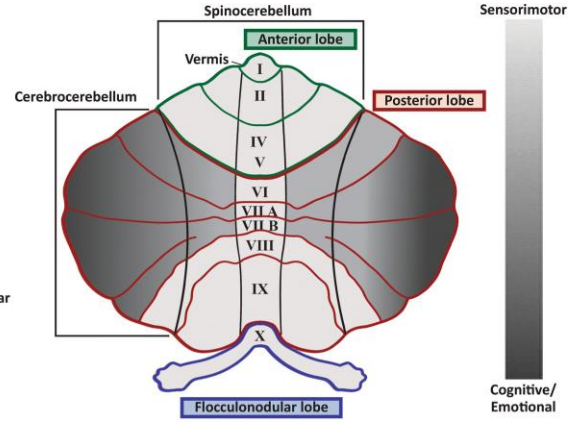
Descriptive anatomy: Internal configuration

Embedded within the white matter are the **deep cerebellar nuclei**, the main output centers of the cerebellum:

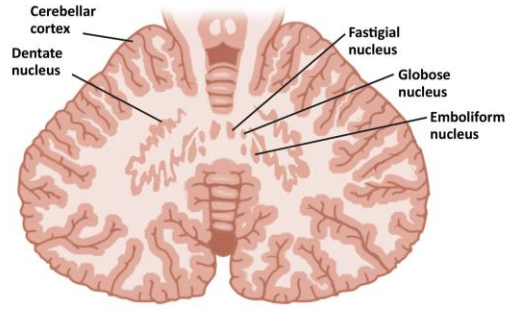
- **Dentate nucleus:** the largest, highly convoluted, associated with planning and coordination of voluntary movements.
- **Emboliform nucleus:** involved in regulation of limb movements.
- **Globose nucleus:**works with the emboliform nucleus as part of the interposed nuclei.
- **Fastigial nucleus** – located medially, linked to balance and posture via vestibular connections.
- These nuclei receive input from Purkinje cells and send output through the **cerebellar peduncles** to other parts of the central nervous system.



Midline sagittal view



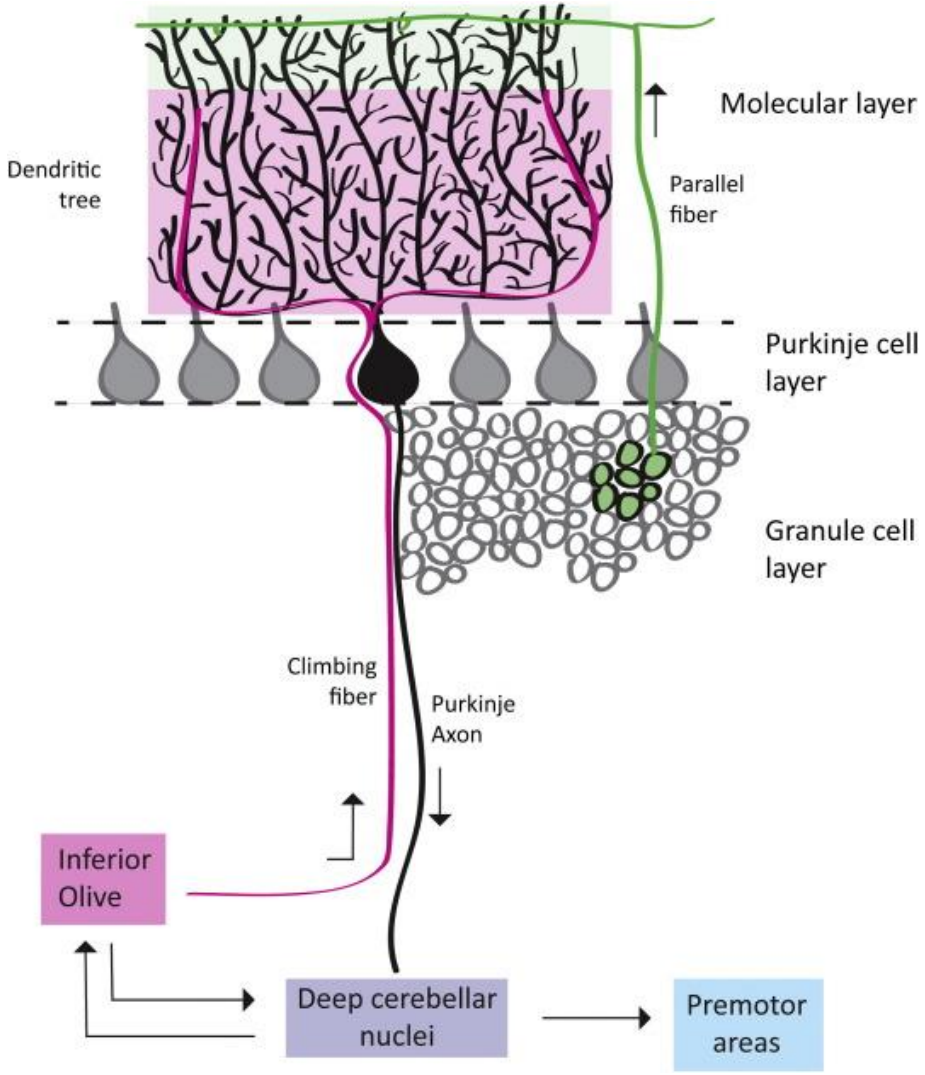
Upper surface (vermis in one plane)

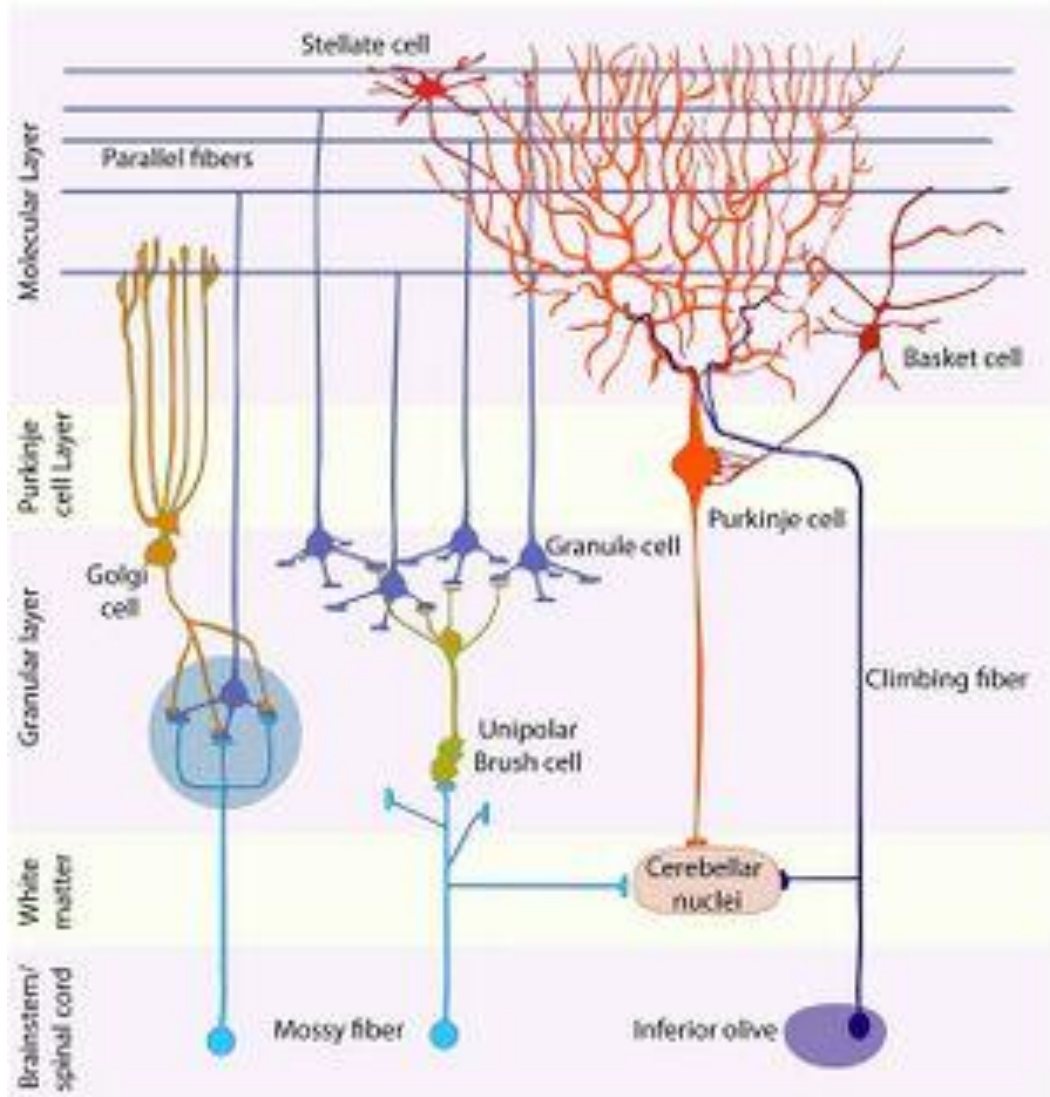


Cross-section showing deep cerebellar nuclei

Descriptive anatomy: Internal configuration

- **The medullary body of the cerebellum** extends through the cerebellar peduncles and beneath the cortex, forming a characteristic branching pattern known as the **“arbor vitae”** (*tree of life*).
- Two main types of afferent fibers are distinguished:
- **Mossy fibers**
- **Climbing fibers**




Stellate cell
Parallel fibers
Molecular Layer
Basket cell
Purkinje cell Layer
Purkinje cell
Granule cell
Golgi cell
Granular layer
Climbing fiber
Unipolar Brush cell
White matter
Cerebellar nuclei
Brainstem/ spinal cord
Mossy fiber
Inferior olive

Descriptive anatomy: Internal configuration

1. **Mossy fibers**

•Originate from multiple sources including the **spinal cord**, **vestibular nuclei**, and **pontine nuclei**.

•They terminate in the **granular layer**, where they synapse with **granule cells** via specialized structures called *glomeruli*.

•Each mossy fiber excites a large number of granule cells, which in turn activate Purkinje cells indirectly through their parallel fibers.

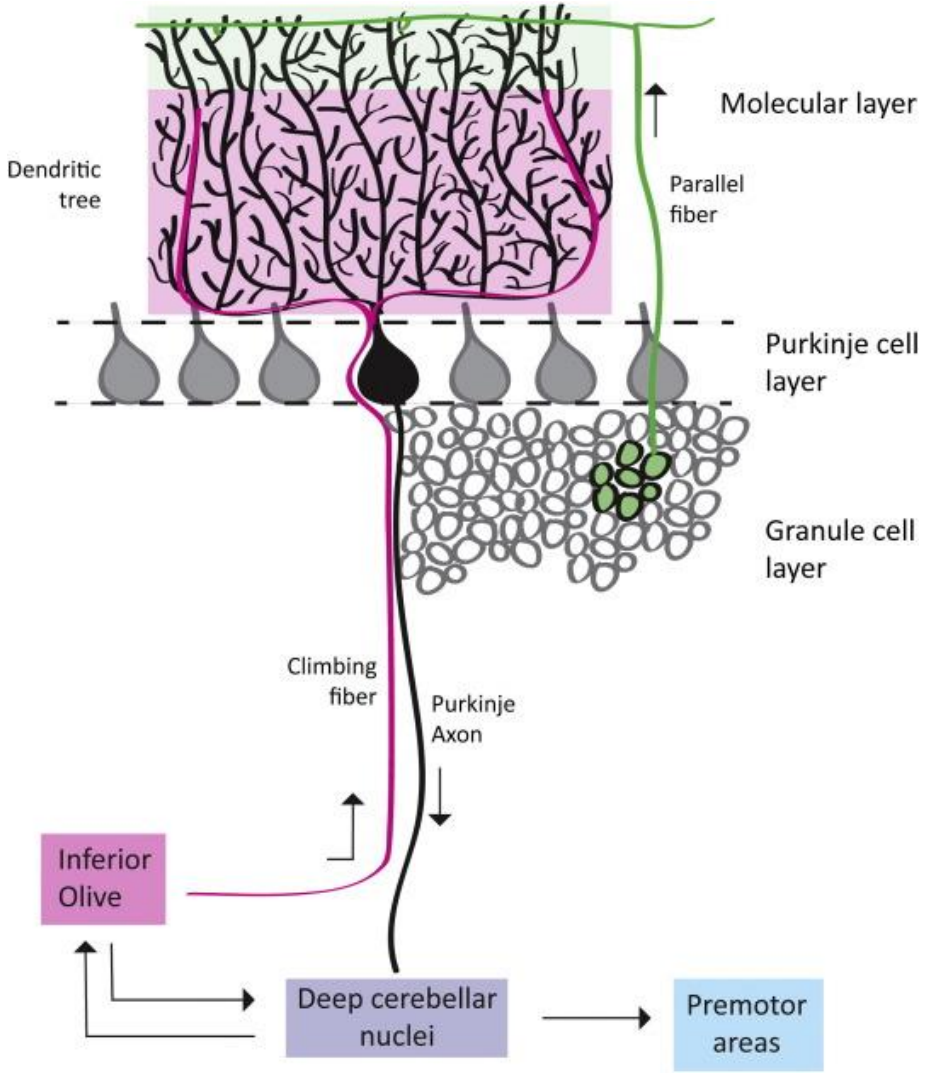
•Their role is to transmit a wide variety of **sensory and motor signals** to the cerebellum.

2. **Climbing fibers**

•Arise exclusively from the **contralateral inferior olivary nucleus**.

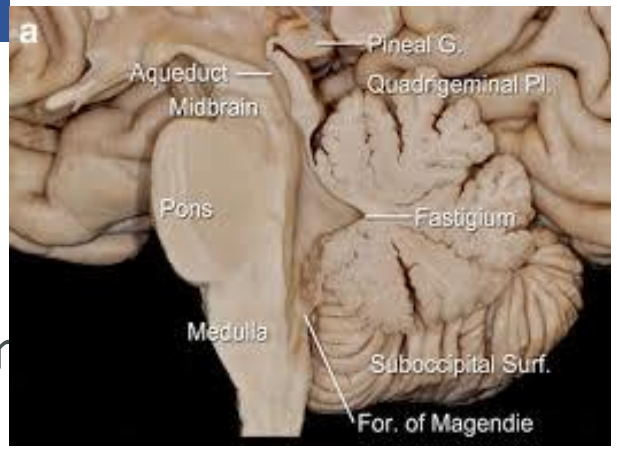
•Each climbing fiber makes powerful excitatory synapses directly onto the dendrites of a single **Purkinje cell**, with hundreds of synaptic contacts.

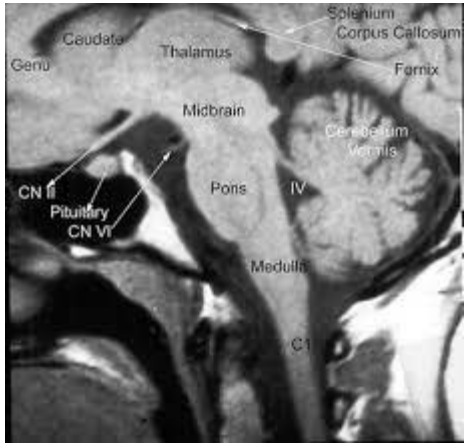
•They are critical for **motor learning**, especially in signaling **errors in movement execution**, allowing the cerebellum to adjust and refine motor commands.



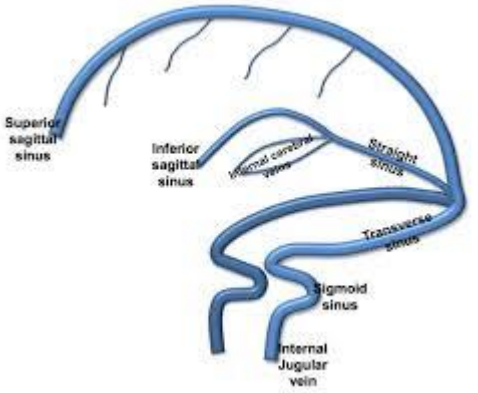
ANATOMICAL RELATIONS

- Superior
  - ➢ the great cerebral vein of Galen
  - ➢ the straight sinus (in the midline of the tentorium cerebelli)
  - ➢ the occipital lobe is also superior to the cerebellum.
- Anterior
  - ➢ the brainstem
  - ➢ the fourth ventricle.
  - ➢ the distal end of the cerebral aqueduct of Sylvius, and the corpora quadrigemini



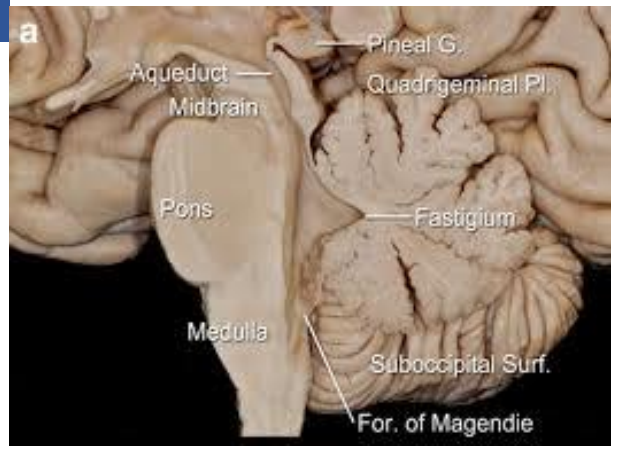


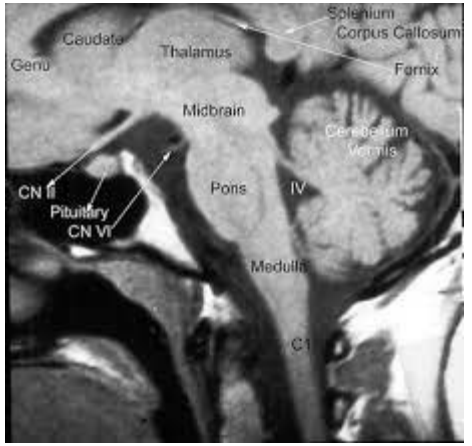




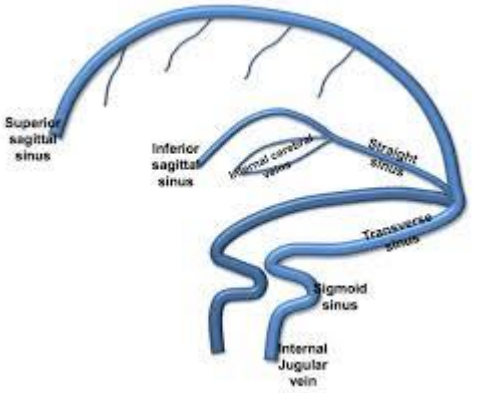
ANATOMICAL RELATIONS

- Downward:
  - ➢ the foramen magnum
  - ➢ the cisterns
- Posterior and lateral
  - ➢ the occipital bone and its overlying layer of dura mater.
  - ➢ The sigmoid sinuses are also found inferiorly
  - ➢ the occipital sinus (if present) and the confluence of sinuses are both posteriorly related.





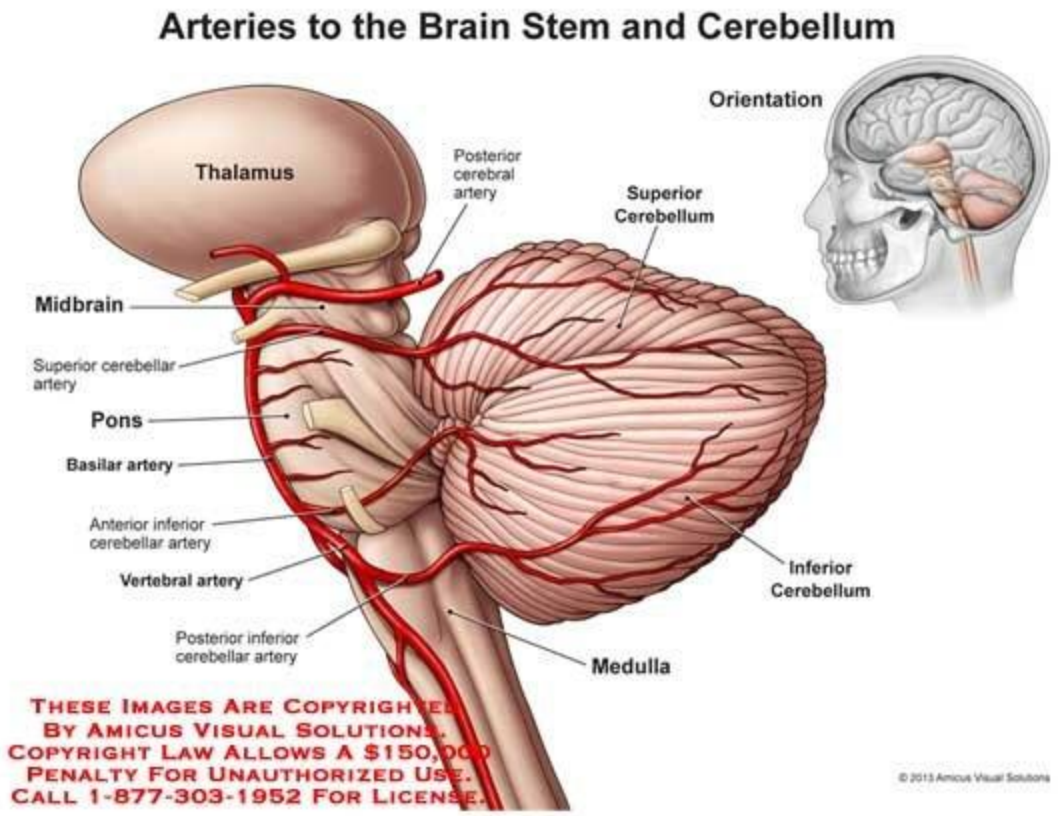




VASCULATURE

The cerebellum receives its blood supply from three paired arteries:

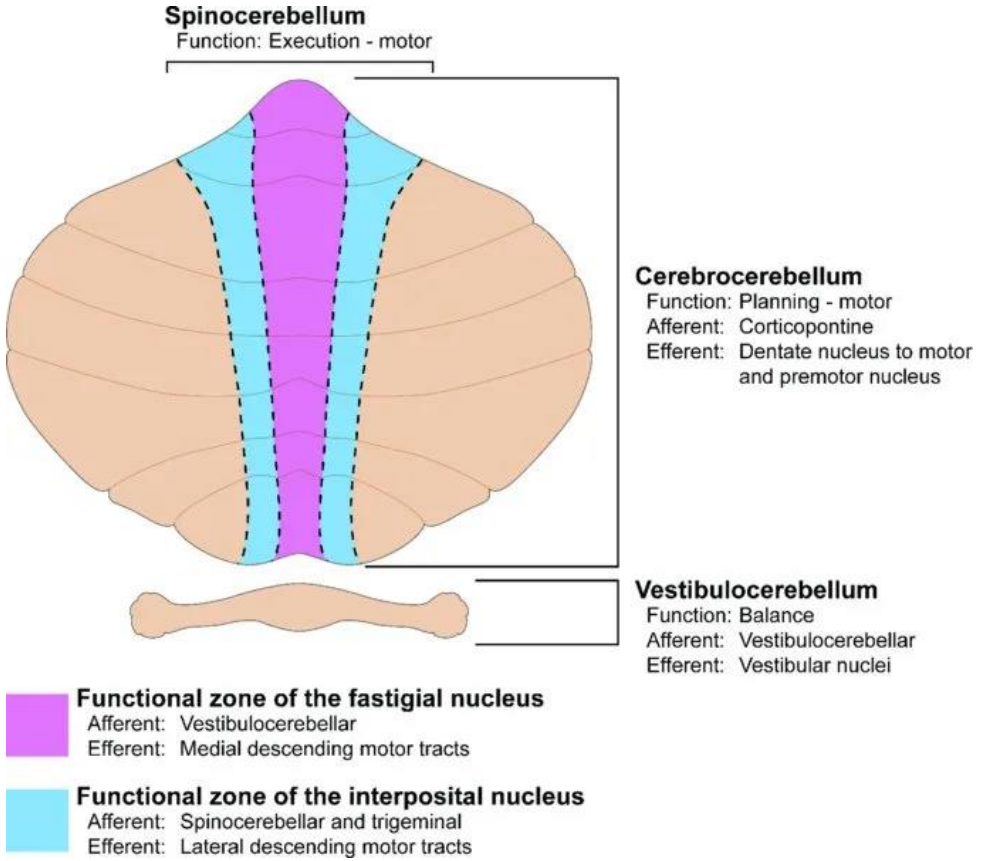
- Superior cerebellar artery (SCA)
- Anterior inferior cerebellar artery (AICA)
- Posterior inferior cerebellar artery (PICA)
- The SCA and AICA are branches of the **basilar artery**, which wraps around the anterior aspect of the pons before reaching the cerebellum. The PICA is a branch of the **vertebral artery**.
- Venous drainage of the cerebellum is by the **superior** and **inferior cerebellar veins**. They drain into the superior petrosal, transverse and straight dural venous sinuses.



Functional anatomy of the Cerebellum

The anatomical subdivisions described above correspond to three major functional subdivisions of the cerebellum.

- ***Vestibulocerebellum.*** The vestibulocerebellum comprises the **flocculonodular lobe** , it is involved in vestibular reflexes (such as the vestibuloocular reflex; see below) and in postural maintenance.
- ***Spinocerebellum.*** a The spinocerebellum comprises the **vermis** and the **intermediate zones** of the cerebellar cortex, as wells the **fastigial** and **interposed nuclei**. It is involved in the integration of sensory input with motor commands to produce adaptive motor coordination.
- ***Cerebrocerebellum.*** The cerebrocerebellum is the largest functional subdivision of the human cerebellum, comprising the **lateral hemispheres** and the **dentate nuclei**. It is involved in the cognitive functions of the cerebellum.



Functional anatomy

Functionally, the cerebellum can be divided into **three major regions**, each with distinct connections and roles:

1. Vestibulocerebellum (Archicerebellum)

1. Corresponds to the **flocculonodular lobe**.
2. Receives input from the **vestibular apparatus and nuclei**.
3. Controls **equilibrium**, **balance**, and **coordinated eye movements**.
4. Main output: to **vestibular nuclei**.

2. Spinocerebellum (Paleocerebellum)

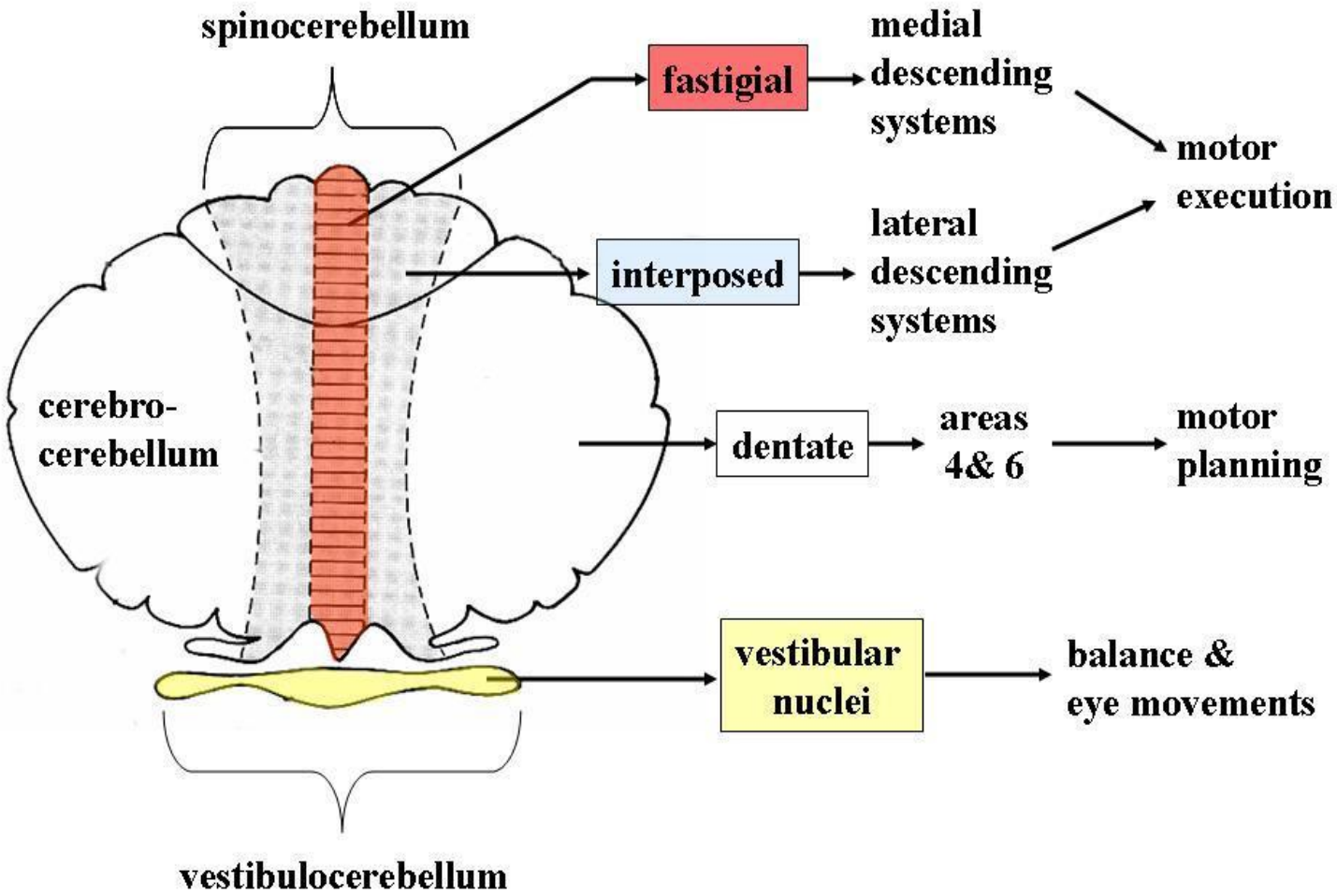
1. Includes the **vermis** (median zone) and the intermediate **zones** of the cerebellar cortex.
2. Receives input from the **spinal cord** (spinocerebellar tracts).
3. Regulates **muscle tone**, **posture**, and **coordination of trunk and limb movements**.
4. Outputs through the **fastigial** and **interposed nuclei**.

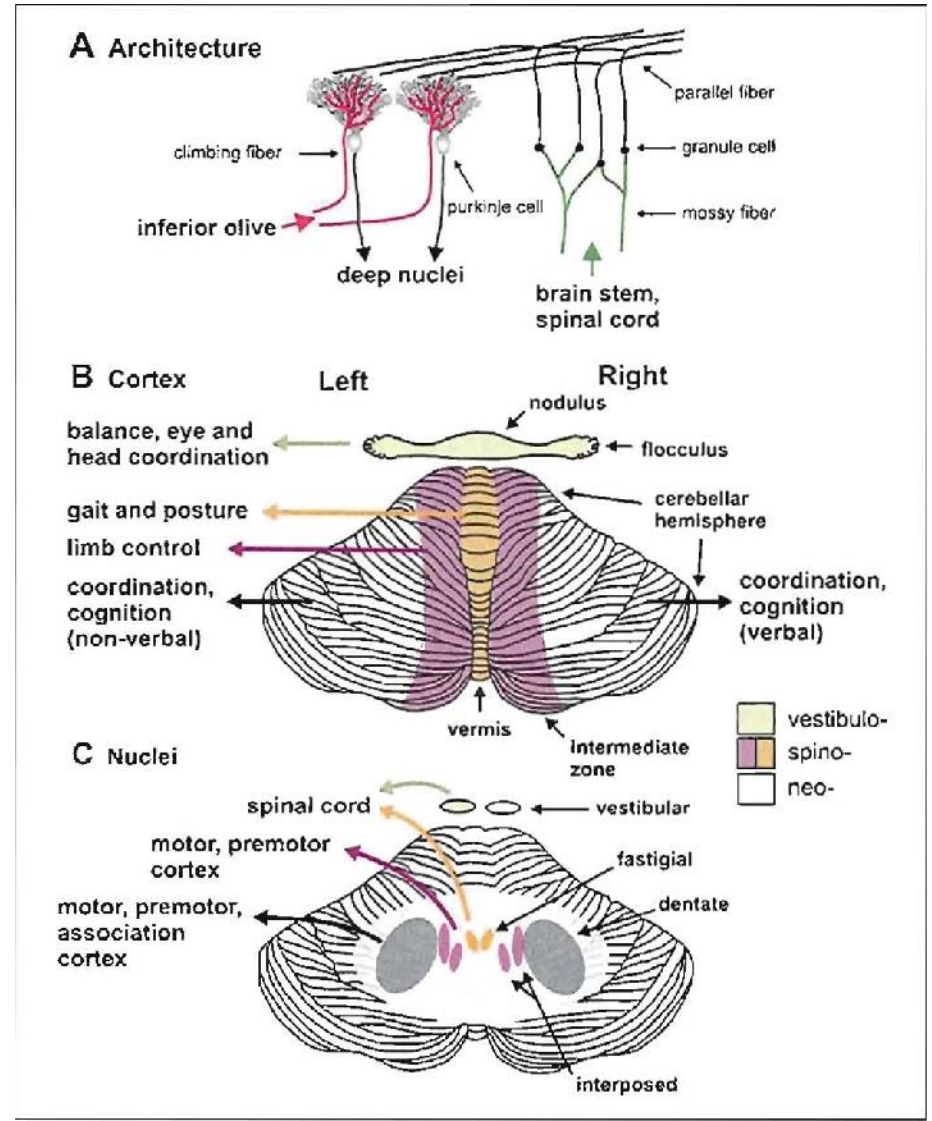
3. Cerebrocerebellum (Neocerebellum)

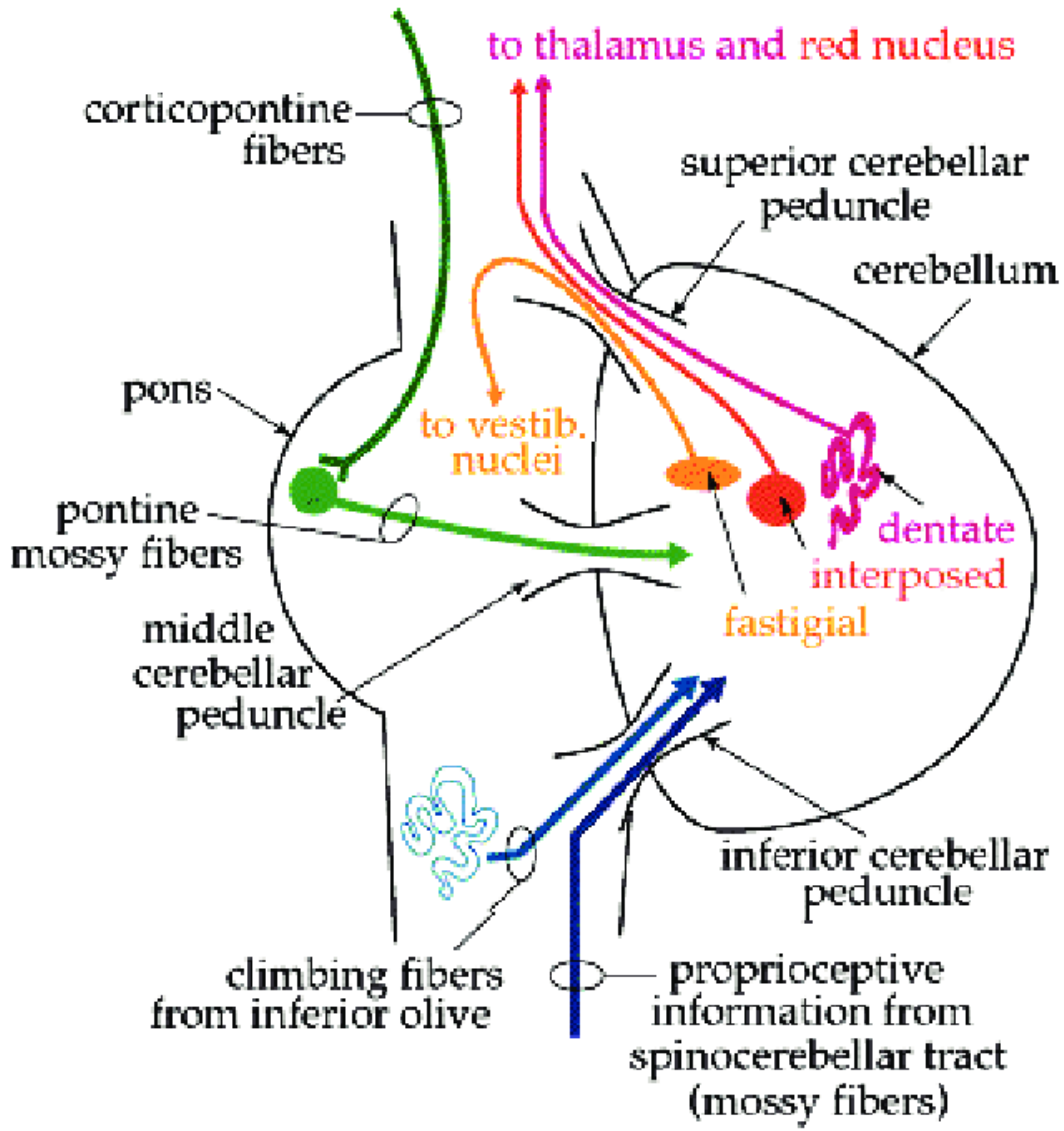
1. Consists of the **lateral hemispheres**.
2. Receives input from the **cerebral cortex** (via pontine nuclei).
3. Involved in **planning, initiation, and timing of skilled voluntary movements**, especially fine motor control.
4. Output through the **dentate nucleus**, projecting to the **thalamus and motor cortex**.

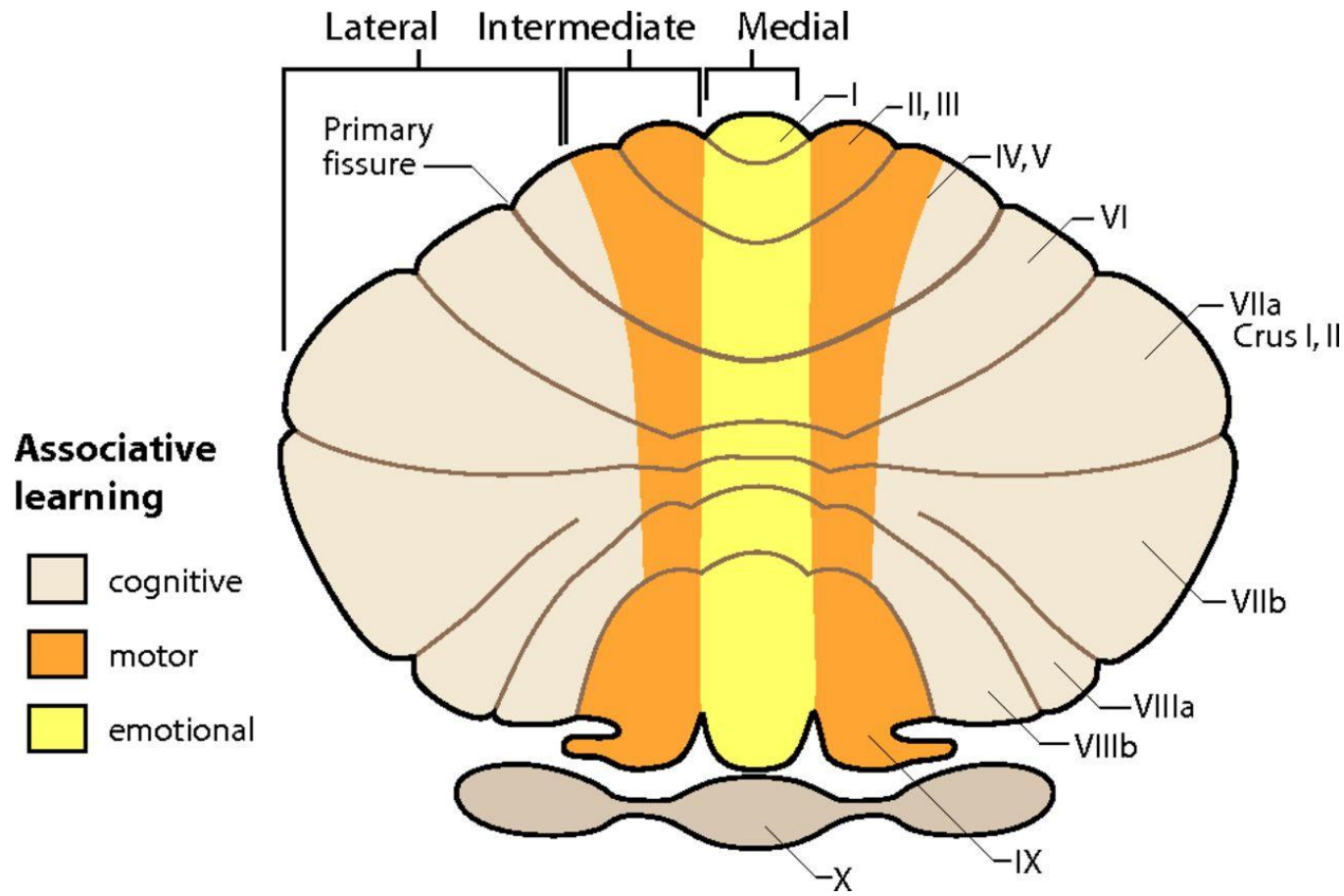
- Together, these regions ensure smooth, coordinated, and precise movements, as well as balance and motor learning.

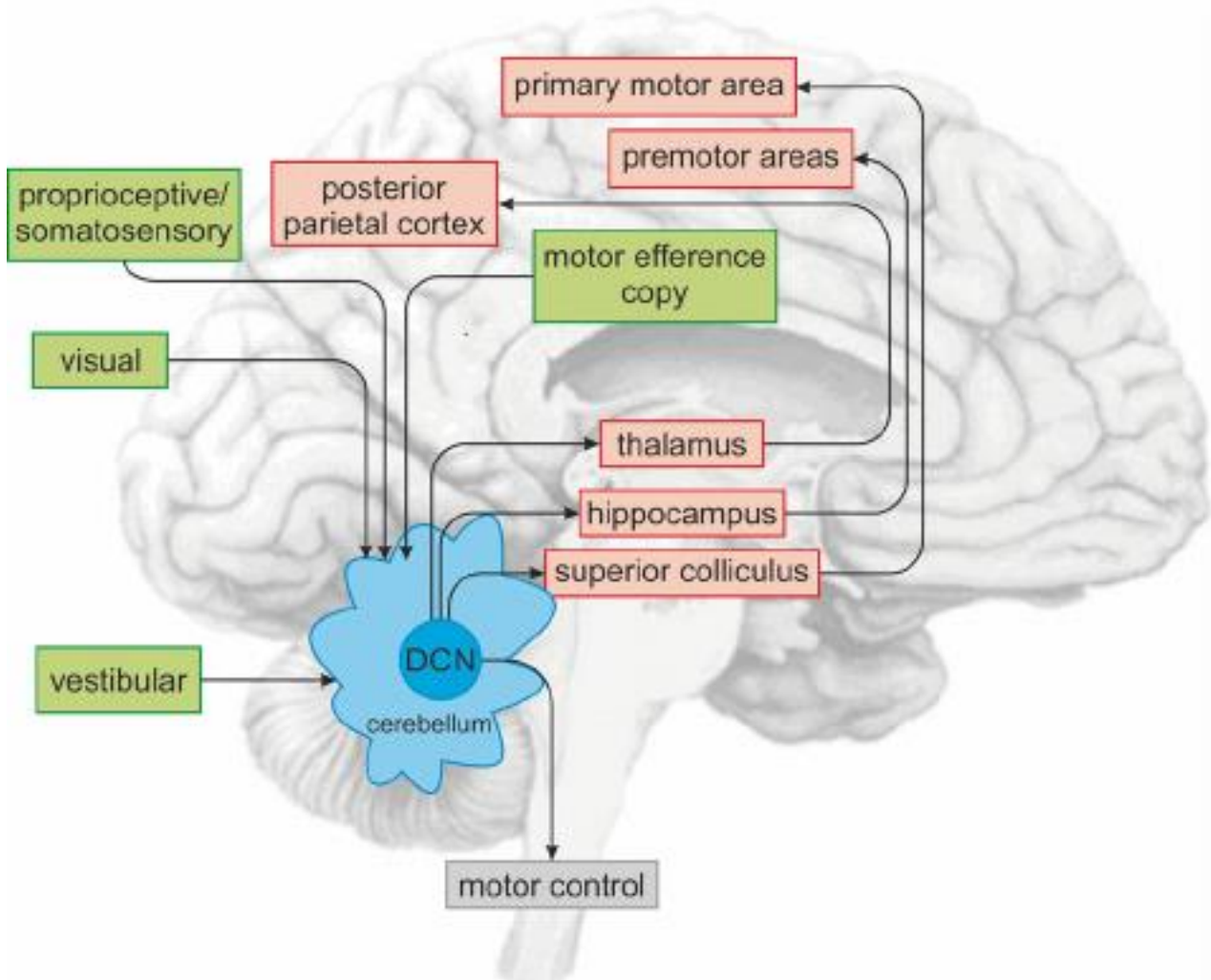
Cerebellar Output



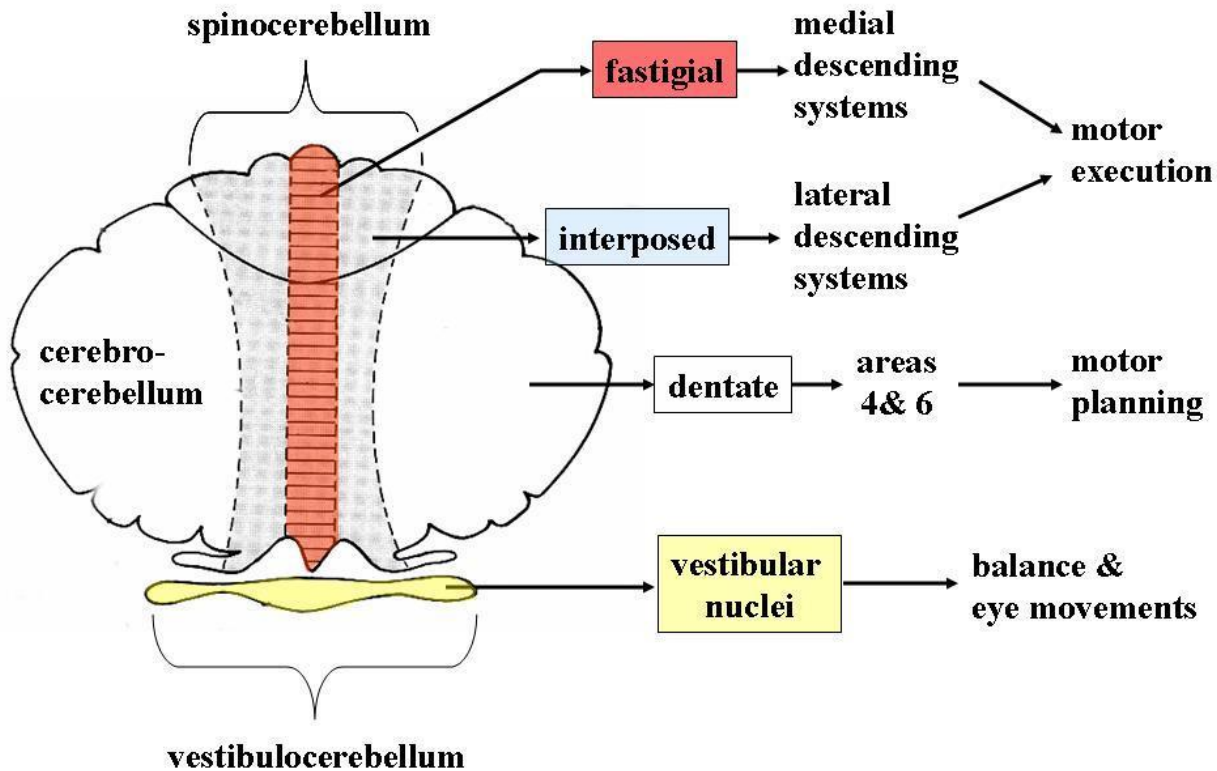


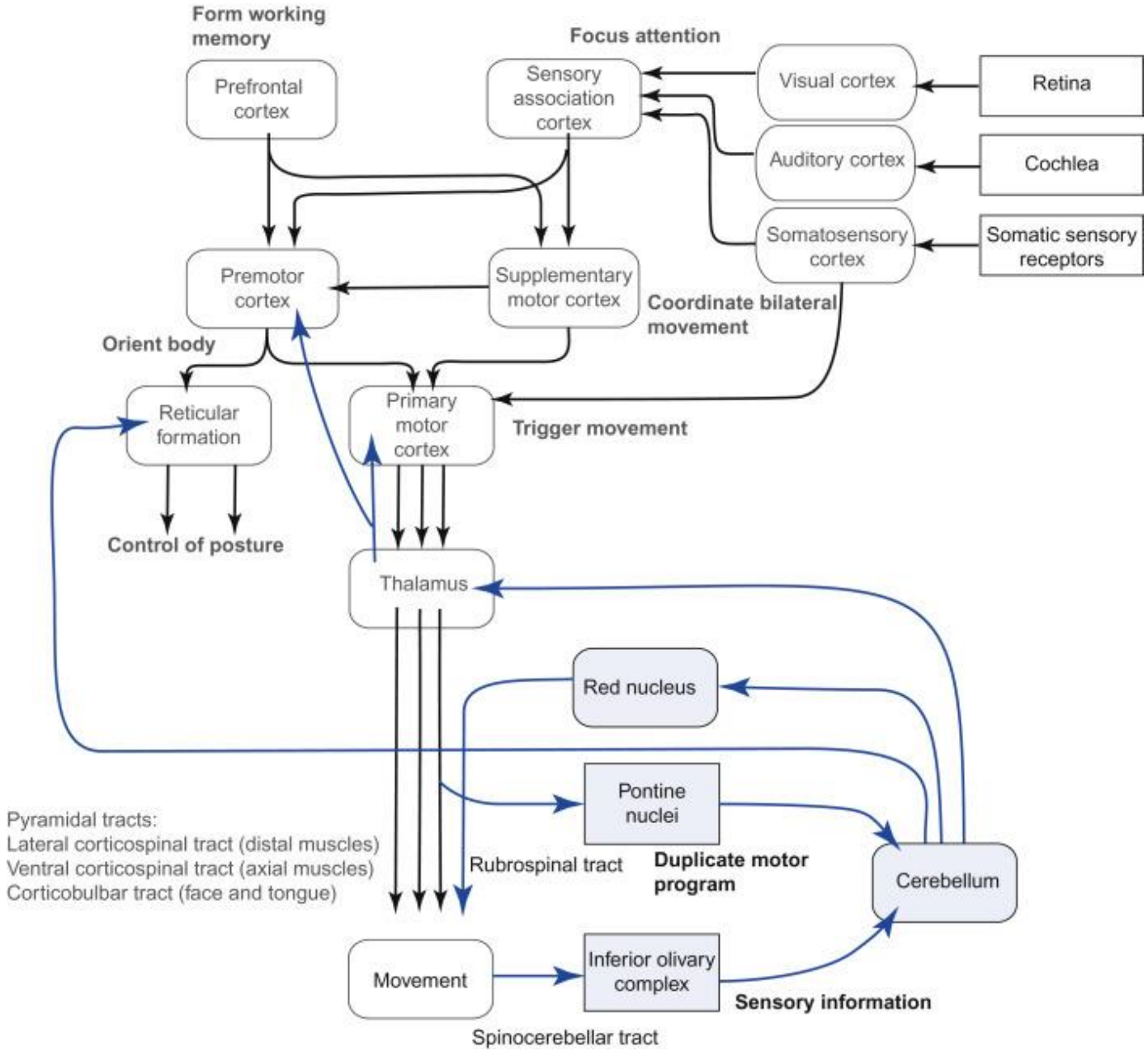

to thalamus and red nucleus
corticopontine fibers
superior cerebellar peduncle
cerebellum
pons
to vestib. nuclei
pontine mossy fibers
dentate
interposed
fastigial
middle cerebellar peduncle
inferior cerebellar peduncle
climbing fibers from inferior olive
proprioceptive information from spinocerebellar tract (mossy fibers)


Lateral
Intermediate
Medial
Primary fissure
I
II, III
IV, V
VI
VIIa Crus I, II
VIIb
VIIIa
VIIIb
IX
X
Associative learning
cognitive
motor
emotional


primary motor area
premotor areas
posterior parietal cortex
proprioceptive/ somatosensory
motor efference copy
visual
thalamus
hippocampus
superior colliculus
DCN
cerebellum
vestibular
motor control

Cerebellar Output




Form working memory
Prefrontal cortex
Focus attention
Sensory association cortex
Visual cortex
Retina
Auditory cortex
Cochlea
Somatosensory cortex
Somatic sensory receptors
Premotor cortex
Supplementary motor cortex
Coordinate bilateral movement
Orient body
Reticular formation
Primary motor cortex
Trigger movement
Control of posture
Thalamus
Red nucleus
Pontine nuclei
Pyramidal tracts:
Lateral corticospinal tract (distal muscles)
Ventral corticospinal tract (axial muscles)
Corticobulbar tract (face and tongue)
Rubrospinal tract
Duplicate motor program
Cerebellum
Movement
Inferior olivary complex
Sensory information
Spinocerebellar tract

Superior Cerebellar Peduncle (Brachium Conjunctivum)

- **Main function:** Primarily an **output pathway** from the cerebellum to other parts of the brain.
- **Connections:**
  - Sends cerebellar output to the **red nucleus** and **thalamus** (which projects to the motor cortex).
- **Role:**
  - Coordinates planned movement, especially fine motor control.
  - Involved in motor learning and adjustments.

Middle Cerebellar Peduncle (Brachium Pontis)

- **Main function: Largest input pathway** to the cerebellum.
- **Connections:**
  - Carries **pontocerebellar fibers** from the contralateral pontine nuclei.
- **Role:**
  - Conveys motor plans from the cerebral cortex to the cerebellum.
  - Helps the cerebellum compare intended movement with actual movement.

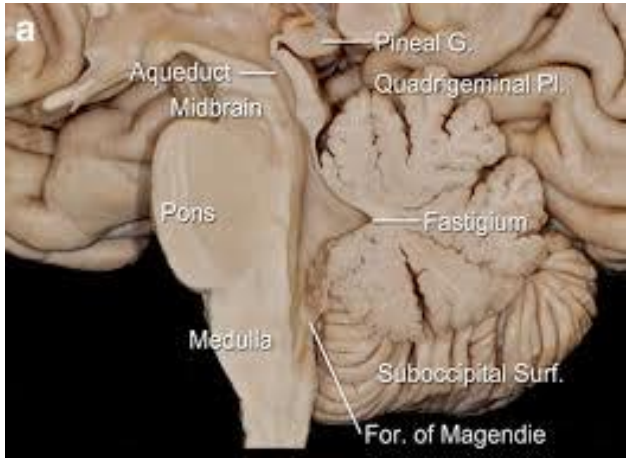
Inferior Cerebellar Peduncle (Restiform Body)

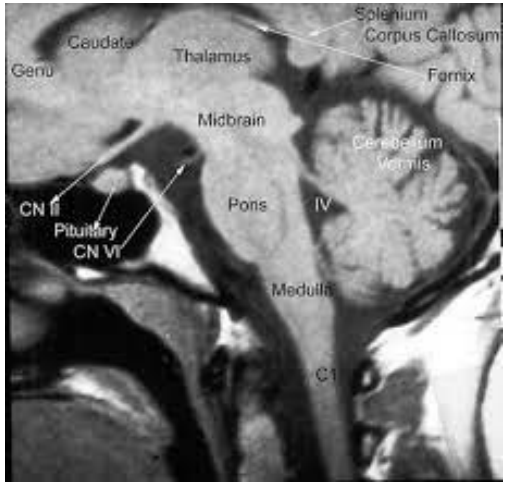
- **Main function:** Carries **both input and output fibers**.
- **Connections:**
  - Inputs from the spinal cord (spinocerebellar tracts) and medulla.
  - Outputs to vestibular nuclei and reticular formation.
- **Role:**
  - Conveys proprioceptive information (body position).
  - Maintains balance and posture.
  - Integrates sensory feedback for motor coordination.

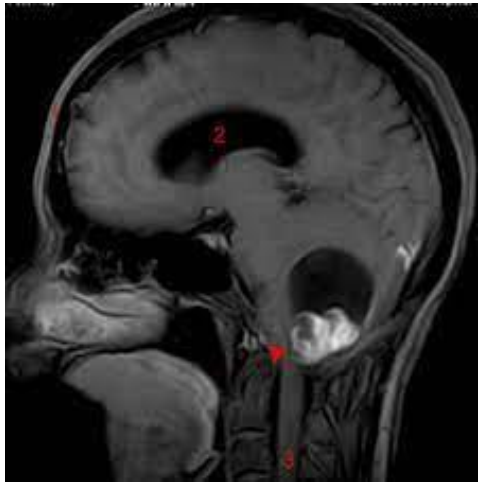
Clinical Significance

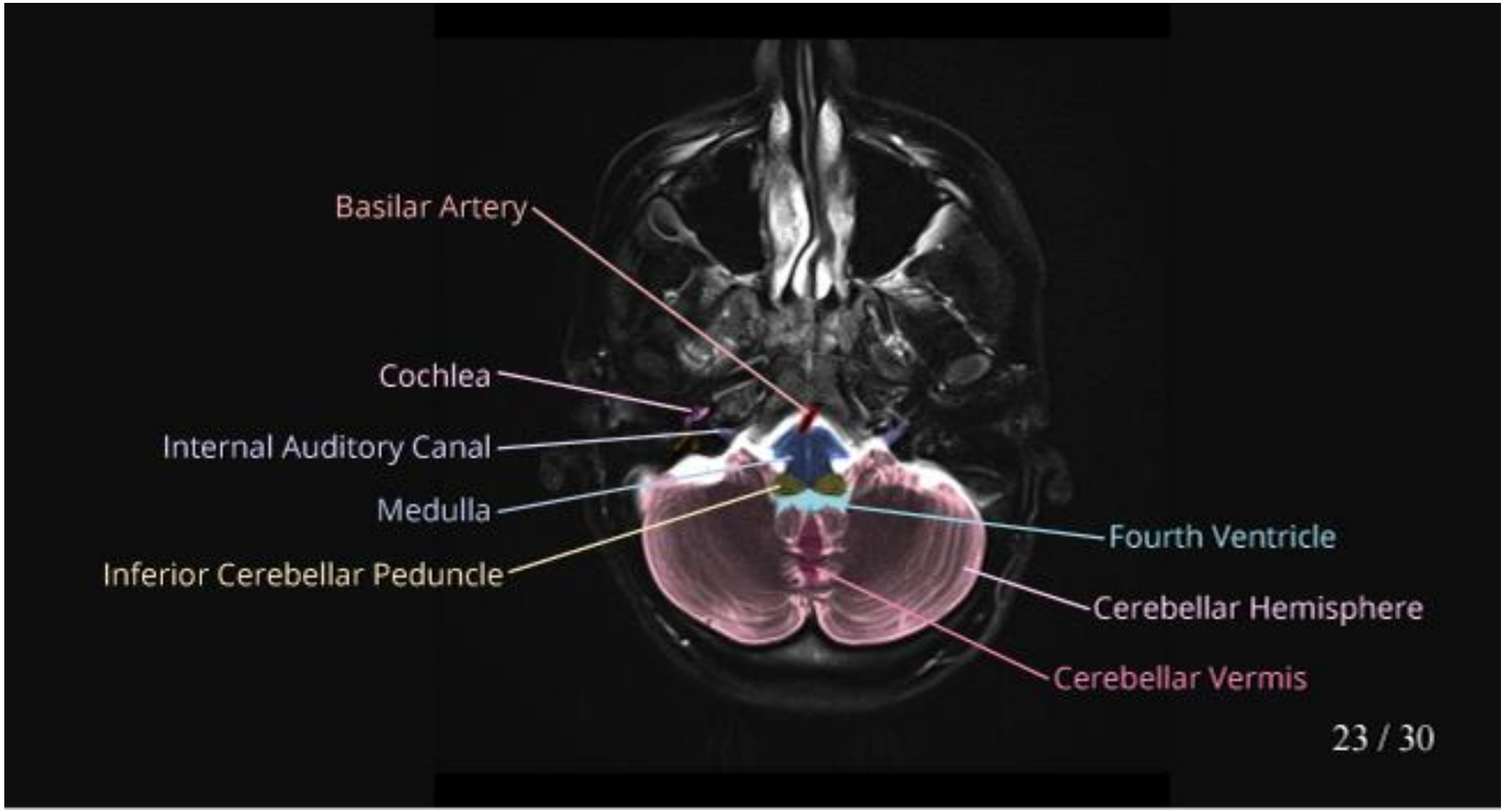
- The cerebellum receives afferent information about voluntary muscle movements from the cerebral cortex and from the muscles, tendons, and joints. It also receives information concerning balance from the vestibular nuclei. Each cerebellar hemisphere controls the same side of the body, thus if damaged the symptoms will occur ipsilaterally.
- Dysfunction of the cerebellum can produce a wide range of symptoms and signs.
- Damage to the flocculonodular lobe (vestibulocerebellum): loss of equilibrium causing an altered walking gait
- Lateral zone damage: problems with skilled voluntary and planned movements leading to errors in intended movements
- Damage to the midline portion: disruption of whole-body movements
- Damage to the upper part of the cerebellum: gait impairments and other problems with leg coordination (ataxia).

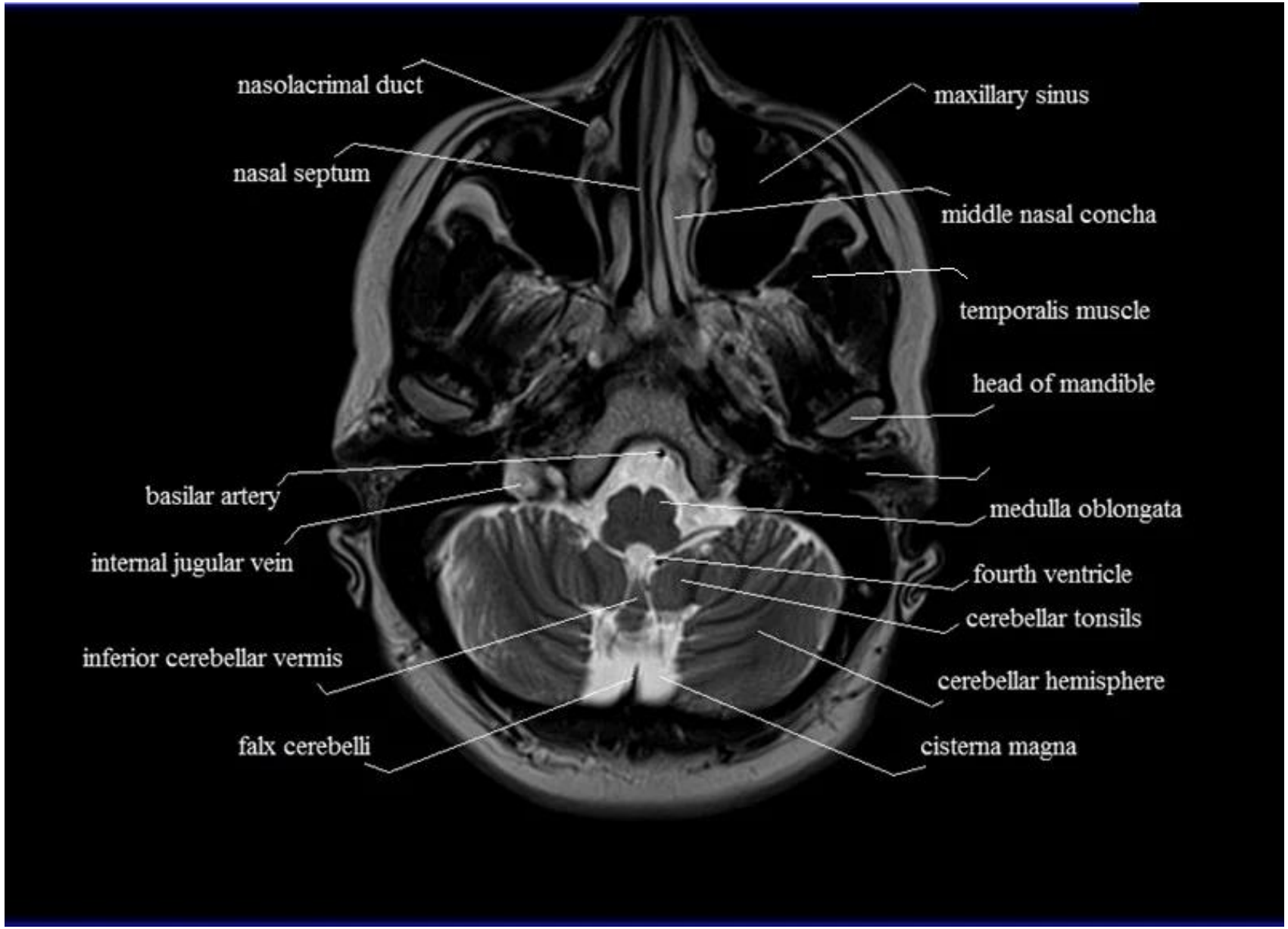
PRACTICAL ASPECTS

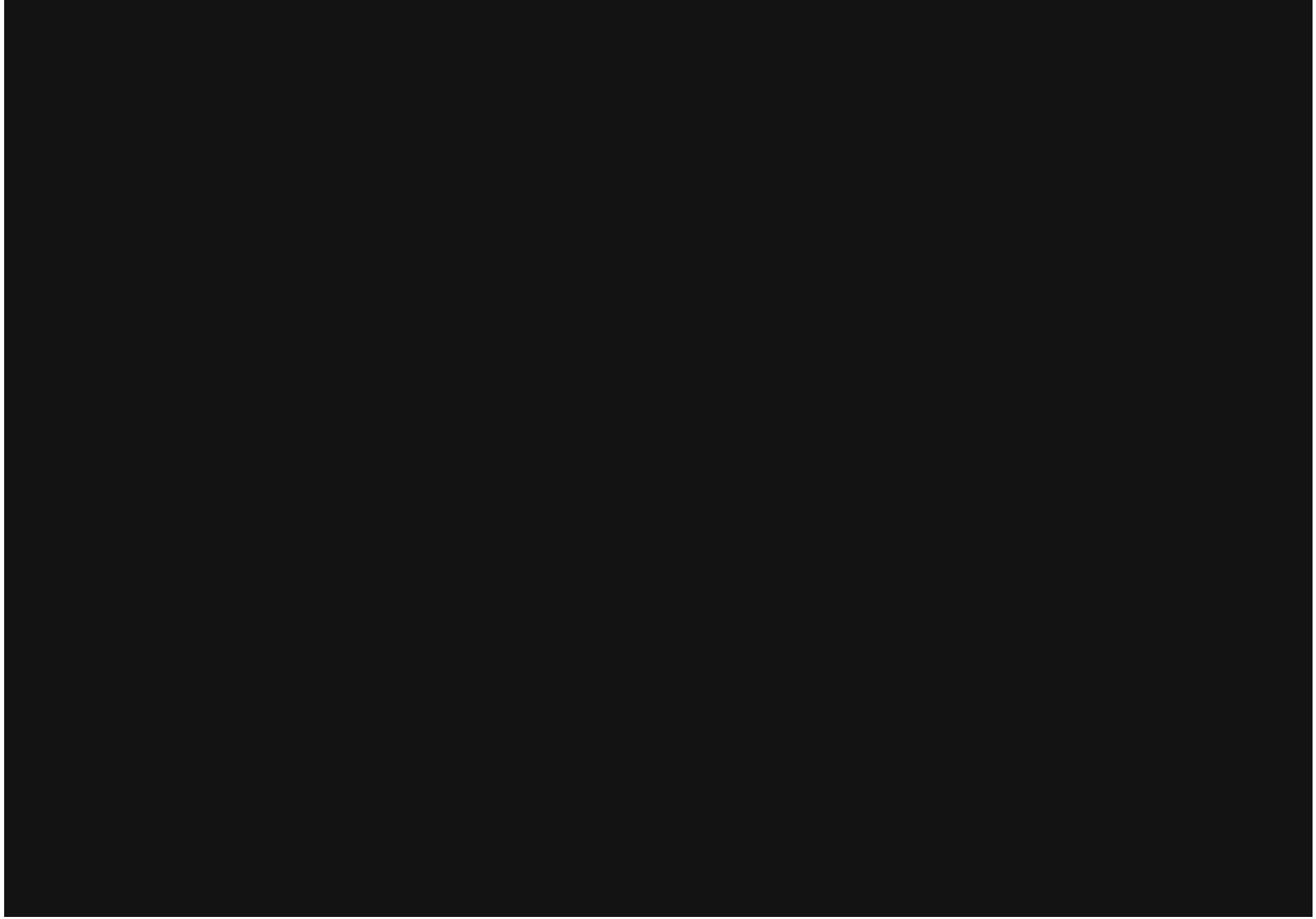







Basilar Artery
Cochlea
Internal Auditory Canal
Medulla
Inferior Cerebellar Peduncle
Fourth Ventricle
Cerebellar Hemisphere
Cerebellar Vermis
23 / 30


nasolacrimal duct
maxillary sinus
nasal septum
middle nasal concha
temporalis muscle
head of mandible
basilar artery
medulla oblongata
internal jugular vein
fourth ventricle
cerebellar tonsils
inferior cerebellar vermis
cerebellar hemisphere
falx cerebelli
cisterna magna



1. Where is the cerebellum located?
A. Anterior cranial fossa
B. Middle cranial fossa
C. Posterior cranial fossa
D. Within the lateral ventricles

2. Which structure connects the two cerebellar hemispheres?
A. Corpus callosum
B. Vermis
C. Dentate nucleus
D. Middle cerebellar peduncle

3. Which fissure separates the anterior and posterior lobes?
A. Longitudinal fissure
B. Posterolateral fissure
C. Primary fissure
D. Central sulcus

4. Which cerebellar peduncle is the largest?
A. Superior
B. Middle
C. Inferior
D. All are equal

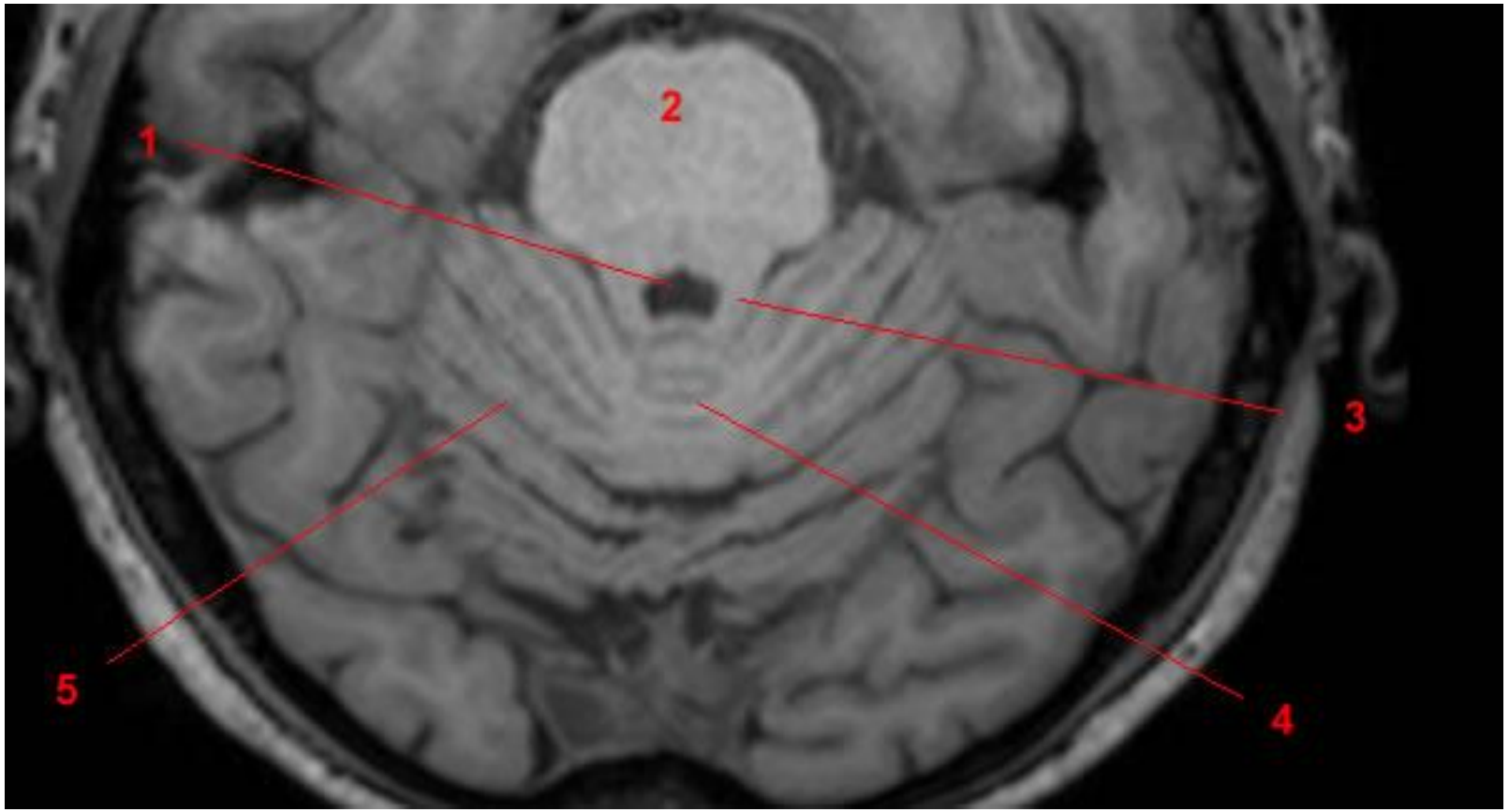
5. Which cells form the principal output of the cerebellar cortex?
A. Granule cells
B. Pyramidal cells
C. Purkinje cells
D. Astrocytes

6. Which deep nucleus is located most laterally?
A. Fastigial
B. Globose
C. Emboliform
D. Dentate

7. Which functional division is most associated with balance and eye movements?
A. Cerebrocerebellum
B. Spinocerebellum
C. Vestibulocerebellum
D. Dentate system only

8. A lesion of the right cerebellar hemisphere most typically causes which finding?
A. Left-sided limb dysmetria
B. Right-sided limb dysmetria
C. Complete bilateral paralysis
D. Isolated loss of pain sensation

•Name the **three functional divisions** of the cerebellum and state their main function.
•List the **three layers of the cerebellar cortex** and the principal cell type in each.
•What is the clinical result of a lesion in the **flocculonodular lobe**?


1
2
3
4
5